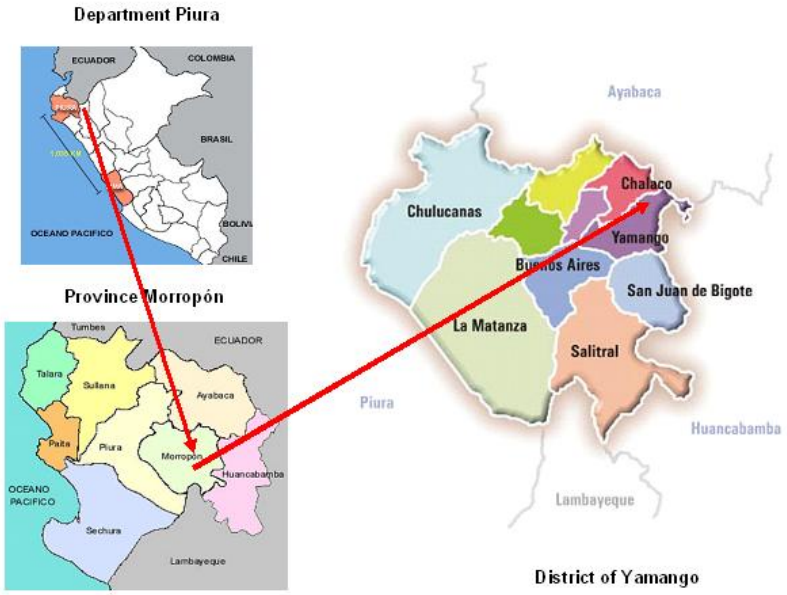
01 Eligibility

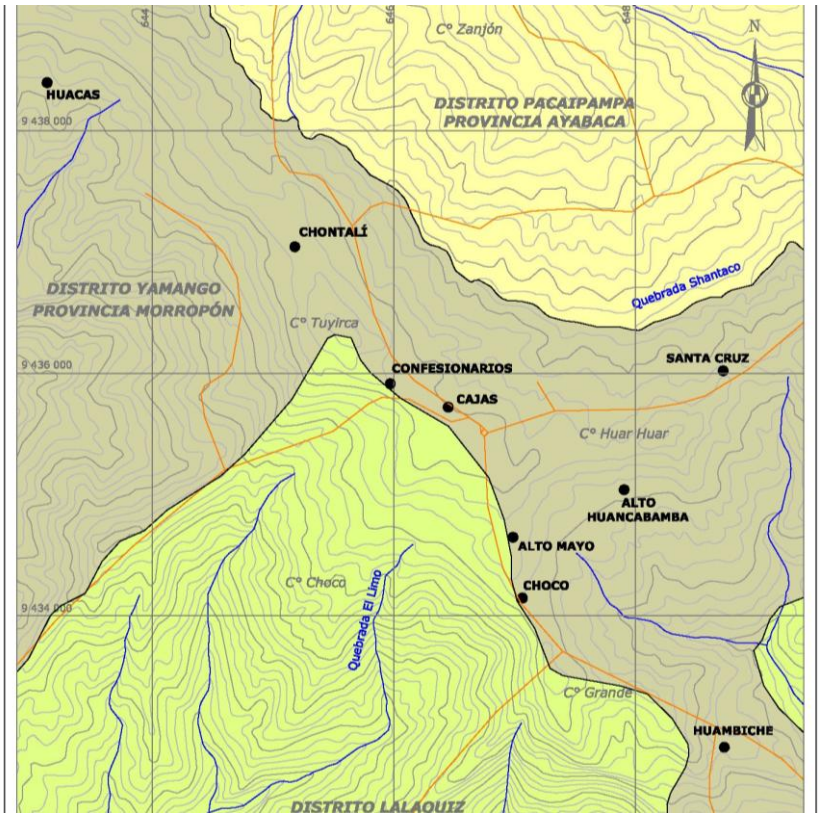
Name of the Project	Reforestation Sierra Piura

1. A description of the historical and the current situation of the project area must be given for the last 50 years. This description must include the development of its socioeconomic situation, its changes in land-uses and changes of property rights.

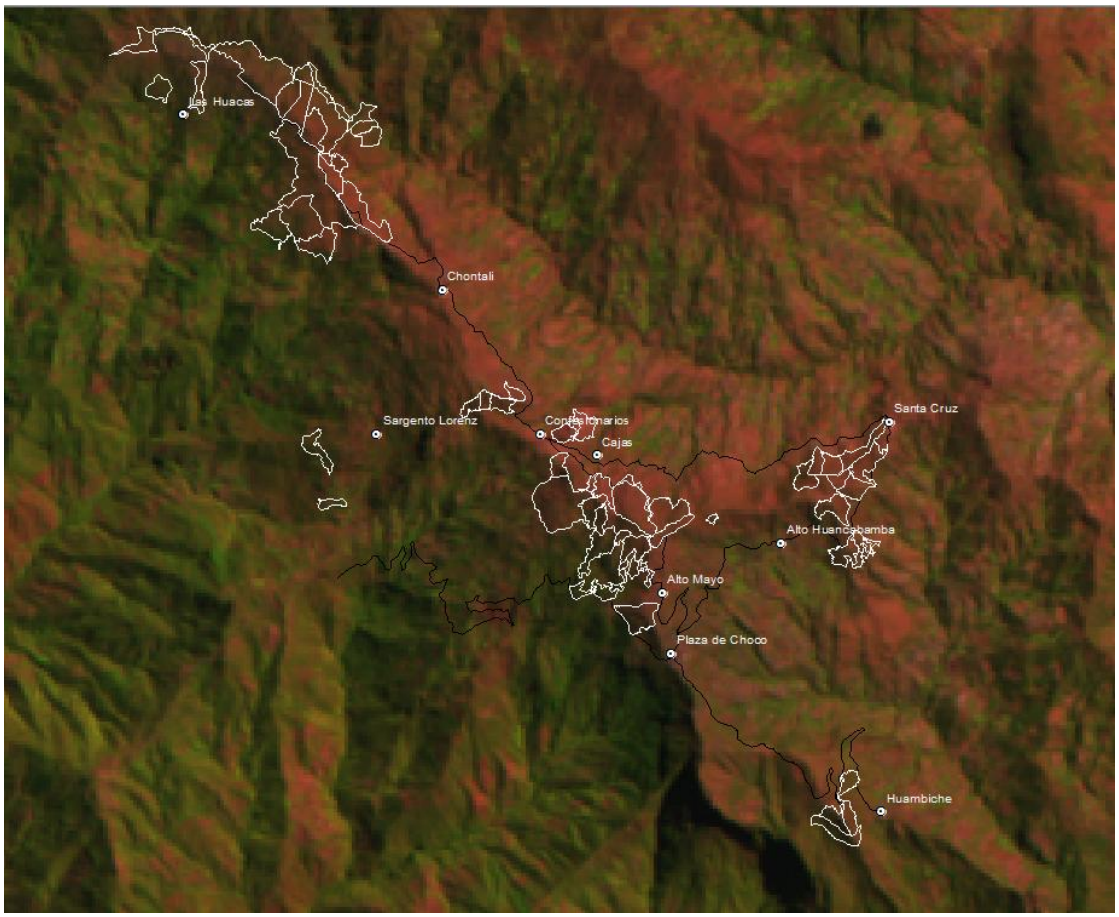
The project is located in the upper part of the Sierra Piura, North Peru. It is being planned on nine communally owned lands around the village Choco, which is part of the community called Yamango, belonging to the district Yamango, province Morropon, department Piura. (See map 1, map 2 and map 3 below).



Map 1: General Location of Project Area



Map 2: Location of the caseríos that participate in the project "Reforestación Sierra de Piura"



Map 3: Location of the caseríos participating in the project Including reforestation areas (white)

In table 1 below, all caseríos are listed that are participating in the project. Also the number of families for each caserío are listed (determined by interviews held in November 2009).

Caserío	N° of families
Choco	105
Huambiche	28
Alto Huancabamba	45
Alto Mayo	28
Santa Cruz	38
Cajas	65
Confesionarios	50
Chontalí	52
Las Huacas	25
Sargento Lorenz	20

Table 1:Names of 9 caserios and nr. of families involved (source: Interviews, Nov 2009)

Degraded grassland/pasture is the only vegetation type that can be found in the plantation area. It is generally assumed (still debated under international scientists) that in the past the 'entire' upper Andes was covered by mountain forest (*Polylepis spp*. etc). The common accepted theory is that due to land pressure and the need for firewood, areas were deforested and have been gradually converted into grasslands. According to interview that were held with local people, deforestation took place some 40-

50 years ago. Garcia, D., et.al., (see ref: 01_01_Garcia_D.etal_2007_DO NOT PUBLISH THIS DOCUMENT.doc, page 2, chapter 1 Introduction, §6) states that political and social events has contributed to the constant deterioration and disappearance of mountain forests in Peru. From the 1960's the Peruvian State started with the construction of roads through the interior of the (high elevation) Andes, which created opportunities to access and cultivate those areas and contributed to the fragmentation of the natural habitats.

The project area is dominated by grasses and some patches of shrubs and low bushes. However, about 1.5 km in the North-West of the project area a native forest remnant is found that is assumed to be representative for the past natural vegetation cover in the project area. This native forest is named 'Bosque Mijal'. Bosque Mijal contains a relative high diversity of woody tree species, shrubs and plants, such as *Alnus jorolensis*, quinual (*polilepis sp*.), cucharillo (Oreocallis grandiflora), Lanche (Myrcia sp.), Chachacomo (Escallonia resinosa), cedro nativo (cedrela sp.).

Supported by NGO's in the '90's (e.g. NGO Villa Nazareth) the farmers in the project area have gained some experience in the reforestation with *Pinus radiata*, *Pinus patula* and to some extent with *Eucalyptus spp* (See photo 1 below).



Photo 1: Forest plantation "Las Huacas" (planted in 1995 with support of NGO Villa Nazareth)

The farmers have also been trained in managing nurseries during that time. Some farmers have established small plots of trees for construction- and firewood.

The main activity in the surroundings of the plantation areas is small scale subsistence farming. Almost all agricultural plots are situated at elevations lower than the locations where the forest plantations are established. The main crops that are cultivated in the surrounding area are: potato (*Solanum tuberosum*), broad bean (*Vicia faba*), wheat (*Tricticum sativum*), barley (*Hordeum sativum*), 'olluco' (*Ullucus tuberosus Caldas*), 'oca' (*Oxalis tuberosa*), maíz (Zea mays). Photo 2 below shows the cultivation areas.



Photo 2: Cultivation areas

In case there is a surplus in the crop production, the crops are being sold at the market. These surpluses are directly related to the use of fertilizers: Farmers that have most surpluses in crop production, are also using most of the fertilizers in the area.
Cattle farming is only done on a small scale due to the low fodder productivity of the (degraded) pastures. The few people that do hold some cattle have an average of 2-3 cows or around 4 sheep. Now and then, farmers produce cheese for the commercial market, which is being sold for about 10 Soles/kg. Sheep meat is being sold for about 80 Soles /11.5kg.



Photo 3: Children herding cattle.

Some small water pools and a few small (non-permanent) river streams supply the project area with water. Electricity and a telephone is only available in caserío Choco. The only First Aid Post in the area is also located in caserío Choco. The surrounding caserios are all depending on the facilities in Choco.

The area has one cemetery that is located in the caserio Confesionarios. All caseríos (except Alto Mayo) have a small school that provides basic level education. Primary schools are only located in the caseríos Choco y Cajas. Secondary education is only offered in caserio Choco (See photo 4: Educative Institute “Juan Bautista” in Choco). According to the Regional Atlas or Piura (see ref: 01_02_AtlasRegional_Piura_2004, page 30 (table), table row nr. 7, Distrito Yamango, Area Rural) around 92.8% of the population of the Yamango district is living below the minimum living standard.



Photo 4:
Educative Institute "Juan Bautista" - Caserío Choco
("Juan Bautista" offers basic, primary and secondary education)

Given the fact that social-economic information about the area is scarce, the 'Atlas Regional de Piura' (see ref: 01_02_AtlasRegional_Piura_2004) has been taken as a reference. This Atlas states that analphabetism is 40%. 67% of the persons older than 15 years did not finish primary education and only 8.3% have finished secondary education. Around 22% of the children between 6 and 12 years old are not going to school. From the children between 13 and 17 years, around 43.8% attend education. An estimated 25% of the children between 9 and 15 years old are behind in education. It is important to emphasize that this information is representative for the situation for the entire district of Yamango, including the urban areas. As such it is estimated that the situation with respect to education is worse in Choco since this population is entirely rural.

The rural community in the area traditionally has a strong social coherence. Construction of houses, irrigation channels and other activities is being realized by an old tradition called Minga (See photo 5: example of Minga), which is a kind of communal social security system: Villagers that need help with certain activities can apply for labor support through this system. The currency that is being used in this system is 'Labor'.



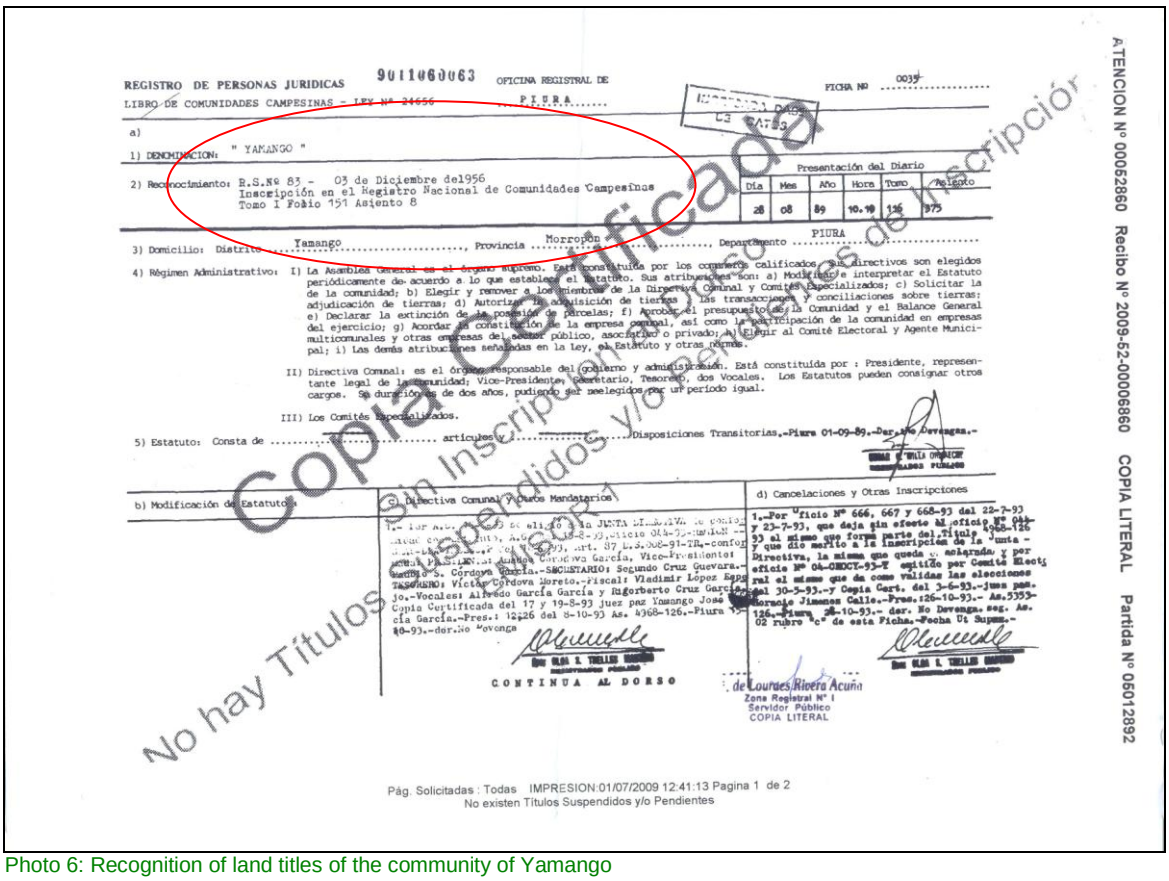
Photo 5. Example of 'Minga' in Chontalí

There are two types of migration in the area: temporal and permanent. Men travel to the lower situated

rainforest to collect wood and come back to the village again after a certain period. This is considered as temporal migration. The second type, permanent migration, deals with youth that migrate to the towns permanently after having finished secondary school. Both types of migration are caused by the lack of employment and the lack of possibilities to improve their living standard.

There is not expectation that this migration pattern will change in the future since till present initiatives to improve living standards of the farmers are not carried out and are neither foreseen.

With respect to land tenure, the community of Yamango has been officially recognized according to 'R.S 83 of 3 December 1956', Law 24656 (see photo 6 below). The community has also been registered in the Public Records:



REGISTRO DE PERSONAS JURIDICAS 9011060063 OFICINA REGISTRAL DE PIURA FICHA Nº 0039

LIBRO DE COMUNIDADES CAMPESINAS - LEY Nº 24656

a)

1) DENOMINACION: " YAMANGO "

2) Reconocimiento: R.S.Nº 83 - 03 de Diciembre del1956
Inscripción en el Registro Nacional de Comunidades Campesinas
Tomo 1 Folio 151 Asiento 8

Presentación del Diario: Día 28 | Mes 08 | Año 89 | Hora 10.10 | Tomo 136 | Asiento 375

3) Domicilio: Distrito Yamango, Provincia Morropon, Departamento PIURA

4) Régimen Administrativo: I) La Asamblea General es el órgano supremo. Está constituida por los comuneros calificados. Sus directivos son elegidos periódicamente de acuerdo a lo que establece el Estatuto. Sus atribuciones son: a) Modificar le interpretar el Estatuto de la comunidad; b) Elegir y remover a los miembros de la Directiva Comunal y Comités Especializados; c) Solicitar la adjudicación de tierras; d) Autorizar la adquisición de tierras. Las transacciones y conciliaciones sobre tierras; e) Declarar la extinción de la posesión de parcelas; f) Aprobar el presupuesto y la Comunidad y el Balance General del ejercicio; g) Acordar la constitución de la empresa comunal, así como la participación de la comunidad en empresas multicomunales y otras empresas del sector público, asociativo o privado; h) Elegir al Comité Electoral y Agente Municipal; i) Las demás atribuciones señaladas en la Ley, el Estatuto y otras normas.

II) Directiva Comunal: es el órgano responsable del gobierno y administración. Está constituida por : Presidente, representante legal de la comunidad; Vice-Presidente, Secretario, Tesorero, dos Vocales. Los Estatutos pueden consignar otros cargos. Su duración es de dos años, pudiendo ser reelegidos por un período igual.

III) Los Comités Especializados.

5) Estatuto: Consta de ..... artículos ..... Disposiciones Transitorias.-Piura 01-09-89.-Der.No Devenga.-

b) Modificación de Estatutos

c) Directiva Comunal y Otros Mandatarios

d) Cancelaciones y Otras Inscripciones

1.-Por Oficio Nº 666, 667 y 668-93 del 22-7-93 y 23-7-93, que deja sin efecto el oficio Nº 044-93 el mismo que forma parte del Título y que dio mérito a la inscripción de la "unta - Directiva, la misma que queda a salvo y por oficio Nº 04-OMDCT-93-T expedido por Comité Electoral el mismo que da como validas las elecciones del 30-5-93.-y Copia Cert. del 3-6-93.-Jues pan. Hernán Jimenes Calle.-Proc.126-10-93.- As.5353-126.-Piura 28-10-93.- der. No Devenga. seg. As. 02 rubro "c" de esta Ficha.-Fecha Ut Supra.-

CONTINUA AL DORSO

Dra. Elsa L. Trelles Ramos
Registradora Pública

Dr. Lourdes Rivera Acuña
Zona Registral Nº I
Servidor Público
COPIA LITERAL

ATENCION Nº 00052860 Recibo Nº 2009-52-00006860 COPIA LITERAL Partida Nº 06012892

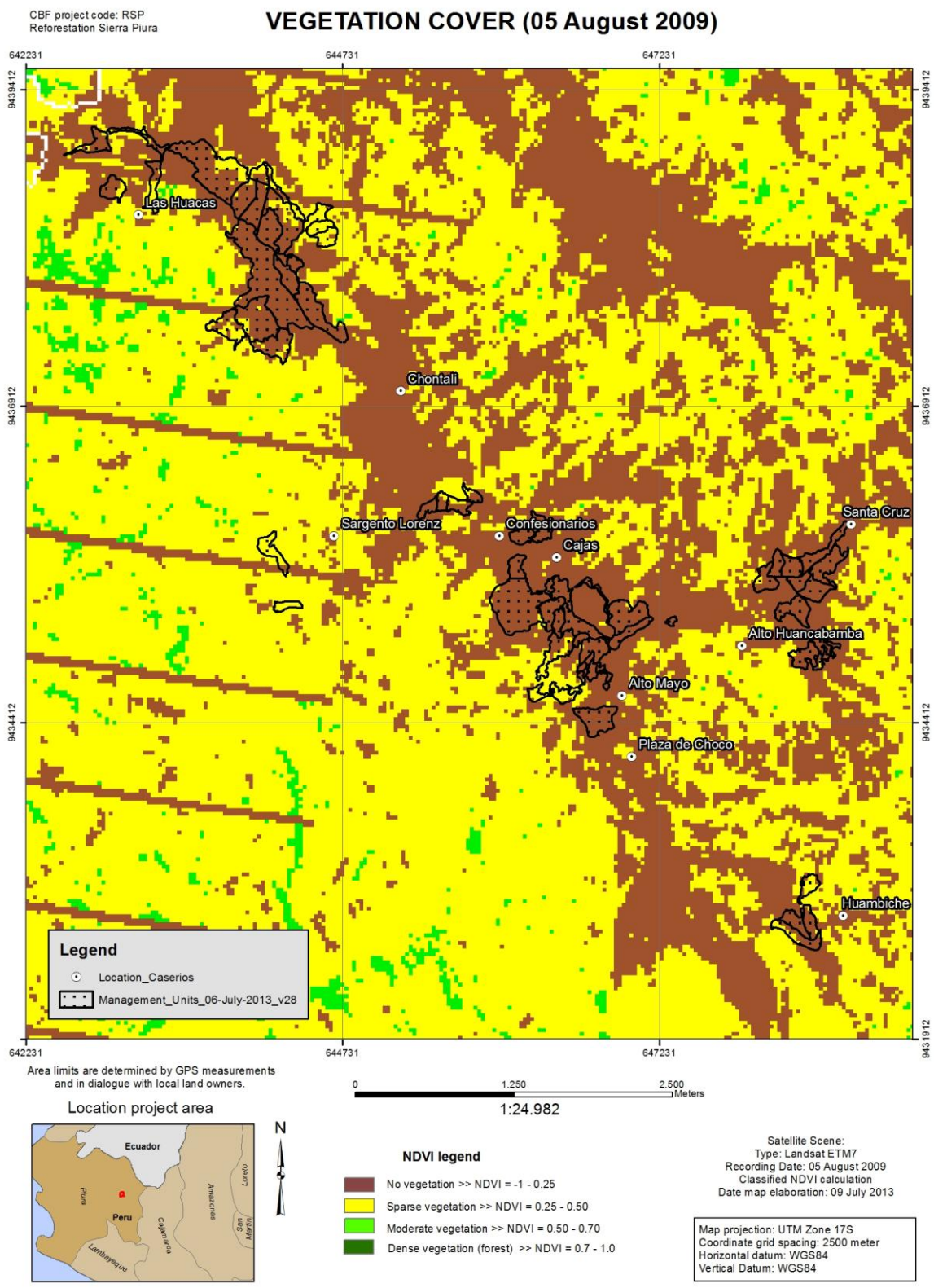
Pág. Solicitadas : Todas IMPRESION:01/07/2009 12:41:13 Pagina 1 de 2
No existen Títulos Suspendidos y/o Pendientes

Copia Certificada
No hay Títulos Sin Inscripción Suspendidos y/o Pendientes

Photo 6: Recognition of land titles of the community of Yamango

For more information regarding land tenure we would like to refer to template 12: Land tenure and CO2.

2. Planting area is ONLY eligible, if the land: a. is not a forest at the project start + b. is planted with trees that result in the creation of a forest + c. has not been forest within 10 years prior to the project start OR has been forest within 10 years prior to the project start and evidence is given that absolutel relation between the project participants and the cause of deforestation exists (e.g. that the f destruction was caused by force majeure) Criterion 2c must be proven by the interpretation of satellite images, aerial photographs, official maps or land-use records.	
a	Deforestation in the project area has taken place in the 1950's. At present (project start), the entire project area is deforested, leaving only some woody plant species and pasture species. This conclusion is supported by the NDVI classification of a Landsat ETM7 satellite scene, recorded on 05 August 2009 (see map 4 below and reference 01_03_A4_Prove_Eligibility_aug2009). The classified Landsat image shows that the plantation areas were entirely deforested in 2009. When we compare the situation in 2009 with the situation in 1998 (see map 6 below, point 2c), we can conclude that deforestation continued from 1998 till present.



Map 4: Limits of the plantation areas plotted on a classified (NDVI) Landsat ETM7 scene, recorded on 05 August 2009.

The reforestation activities will be carried out on these deforested and degraded pasture lands. No other agricultural activity is taken place in the project area besides very extensive cattle farming (only a small number of cows and sheep are being held (up to 4 animals per farmer) by

only a few farmers). Photo 1 and 2 below that were taken at the project start (November 2009) show that the project area is entirely deforested (no presence of woody species).



Photo 1: Current vegetation in the project area.



Photo 2: Current vegetation in the project area.

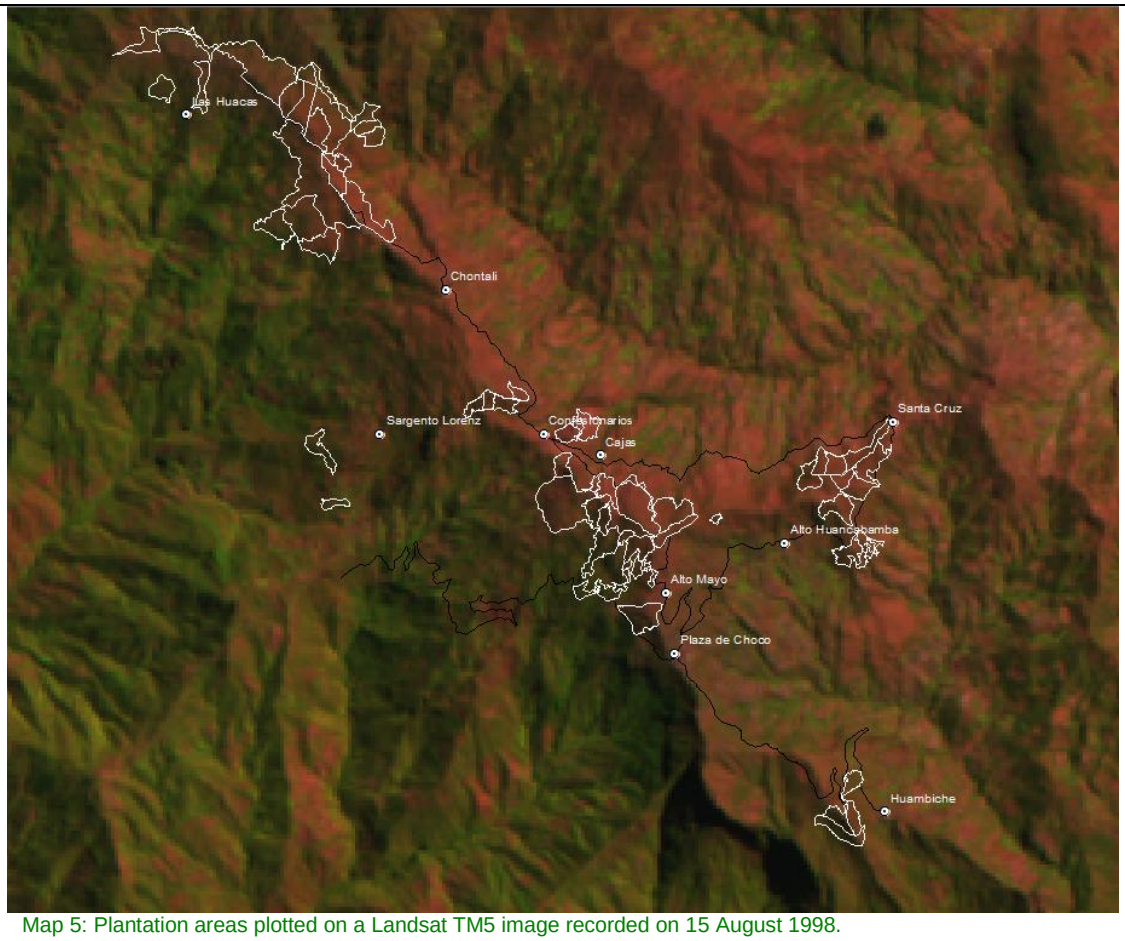
b.

The following four tree species will be planted in the plantation areas:

1. *Pinus patula*
2. *Pinus radiata*
3. *Alnus jorullensis*
4. *Polylepis incana*

Around 13 areas have been appointed by the local farmers (project stakeholders) to become reforested. These areas are subdivided in 69 management units. Although some management units are fairly small, merged the areas are each larger than 0.5ha. The tree species will be systematically planted at a distance of 3 x 3m (1111 trees/ha) or 2.6 x 2.6m (1480 trees/ha). By doing so, the vegetation cover will be >30%. Furthermore, it is expected under normal growing conditions, the trees will reach a height of at least 5m. Eventually in about 10-15 years after the first planting activities, under normal growing conditions and with proper forest management, a plantation forest will be established that meet the criteria of the DNA of Peru:

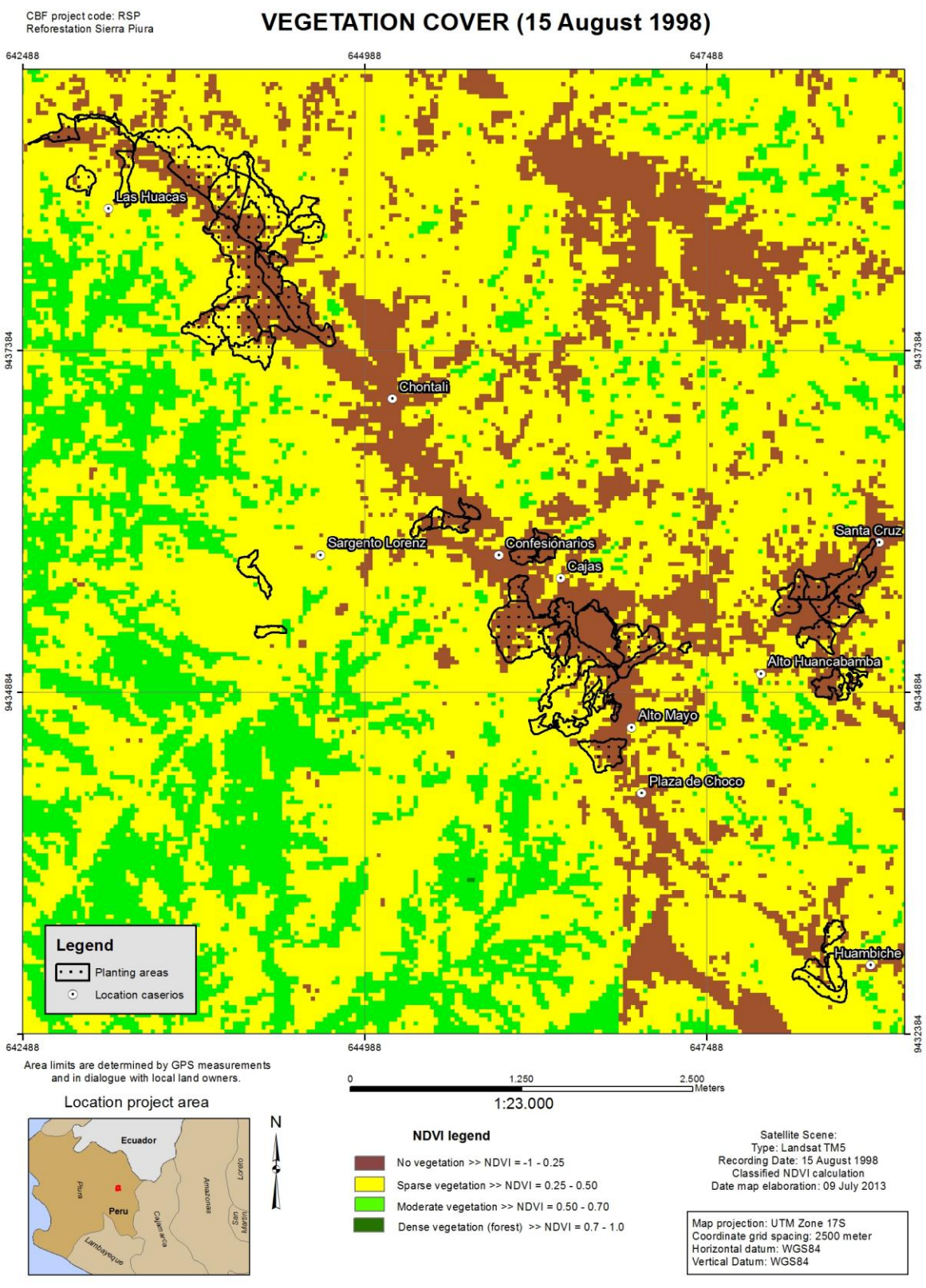
	According to the definition of the Designated National Authority (DNA) of Peru, a forest must reach under the in-situ activities all of the following criteria: • a land area of minimum 0.5 ha • a tree crown cover of minimum 30 % • a mean tree height of minimum 5.0 m (Source: unfccc web-site)
c	Below a map (map 5) is shown of with the outer limits of the plantation areas (consisting of 68 management units) plus the locations of the involved caserios. The plantation areas are plotted on a Landsat TM5 image that was recorded on 15 August 1998.



Map 5: Plantation areas plotted on a Landsat TM5 image recorded on 15 August 1998.

The Landsat TM5 image in map 5 has a band combination 5/4/3 (RGB). This band combination shows healthy forested areas in fresh green colors while bare soils are mauve (source: http://web.pdx.edu/~emch/ip1/bandcombinations.html). Map 5 indicates that most of the plantation areas are situated in areas with a mauve color, which means that these areas were already deforested in August 1998.

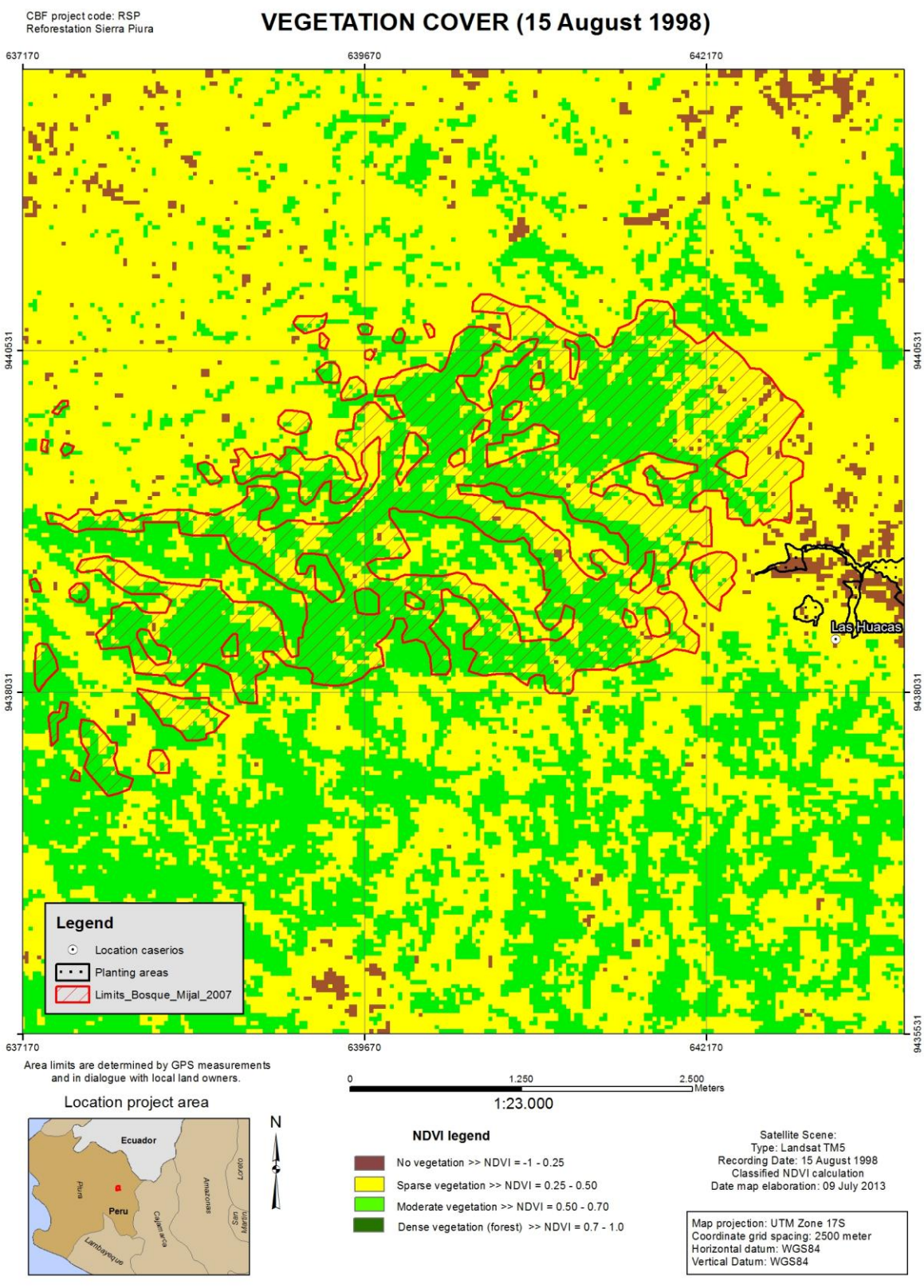
To support the prove of deforestation in 1998, NDVI values were calculated (see also ref: 01_04_A4_Prove_Eligibility_aug1998) for a Landsat TM5 image, recorded on 15 August 1998. First, the Landsat image was atmospherically corrected (dark pixel substraction method). After that, NDVI values for the image were calculated (NDVI = ([NIR-RED] / [NIR + RED]). Subsequently the NDVI image was classified according to the threshold values that are shown in the upper right map legend.
Map 6 below, shows the result of the calculated NDVI values (recorded on 15 August 1998).



Map 6: Limits of the plantation areas plotted on a classified (NDVI) Landsat TM5 scene, recorded on 15 August 1998.

The NDVI map was validated by means of a neighboring native forest remnant called 'Bosque Mijal'. This native forest remnant is located about 1.5 km North-West of the plantation areas and is

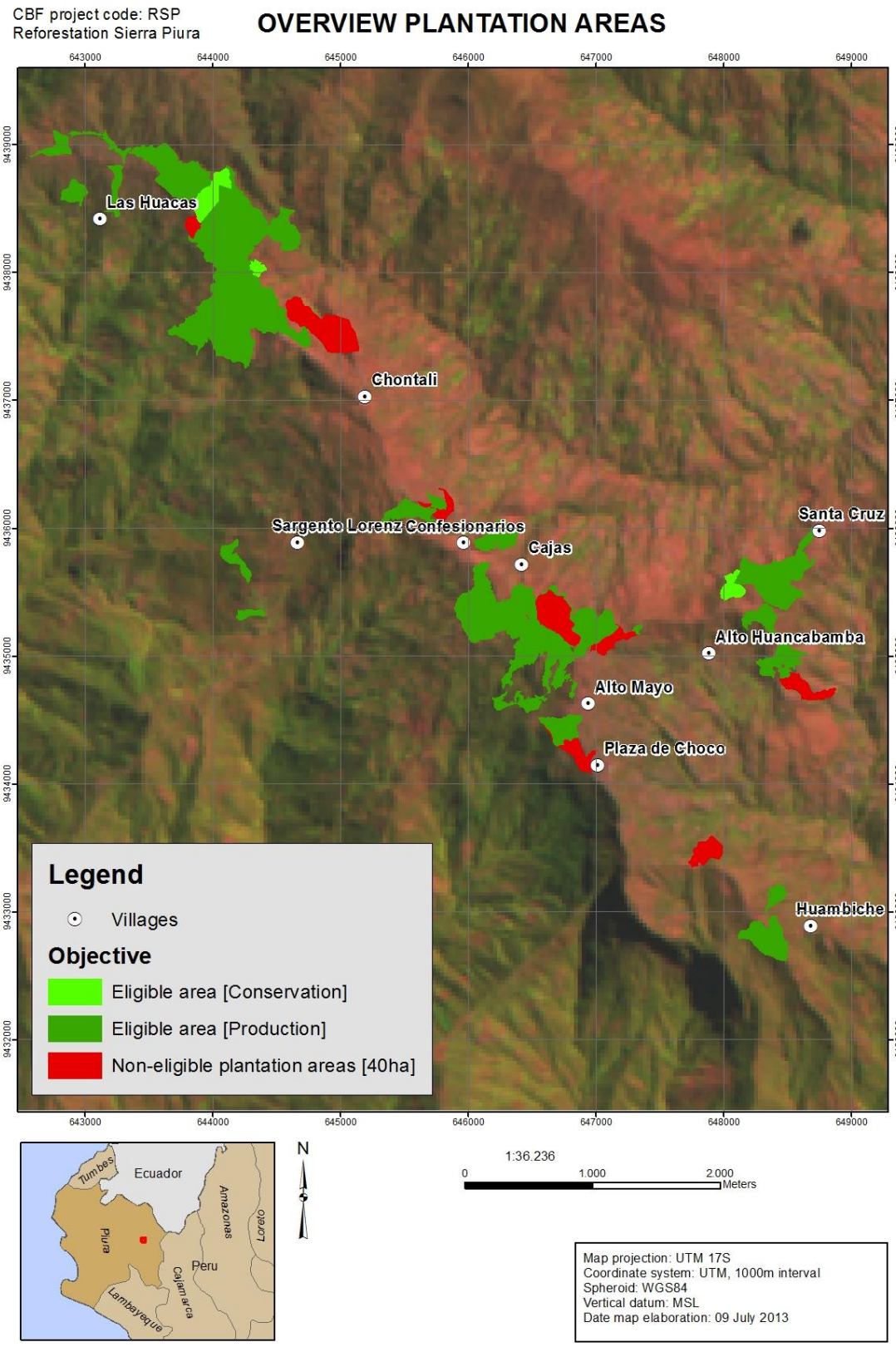
	still relatively well preserved. Bosque Mijal consist of various ecotypes, varying by changes in elevation. Map 7 below shows the natural limits of Bosque Mijal (red delineated areas) plotted on the same classified NDVI scene (Landsat TM5 of 15 August 1998). As can be seen in map 7, Bosque Mijal is almost entirely classified as 'Moderately vegetated'. This proves that the Landsat TM5 scene is correctly classified and thus we can assume that the project areas indeed were deforested in August 1998.



Map 7: Limits of Bosque Mijal (2007) plotted on a classified (NDVI) Landsat TM5 scene, recorded on 15 August 1998.

On the project's website (www.climateprojects.info) short videos are uploaded wherein project participants (farmers) are interviewed. The videos give an impression of the present vegetation in the foreseen plantation areas. The analysis with band combination 5/4/3, the classified (NDVI)

	Landsat scene and the interviews held with the farmers (see www.climateprojects.info), support the conclusions that the plantation areas were already deforested ten year ago. The project area does not contain areas that are not conform with the eligibility criteria. However, due to the fact that in the without-project scenario, around 38 ha (1.5 ha/y) would be planted by the farmers in a 25 year period, 40ha of plantation area has been subtracted from the total plantation area (253ha), resulting in a total plantation area of 213ha. Map 8 below (see also 01_05_A4_Eligibility_Map_with_areas.pdf) provides an overview of the division of the project area in eligible area, eligible plantation areas and eligible conservation areas. We emphasize that the conservation areas are also plantation areas because the conservation areas will be planted with *Polylepis incana*. These areas however, will not be harvested for production timber. The Polylepis plantations will only receive management in case forest development and forest health threatens to stagnate.



Map 8: Division of project area in eligible production area, eligible conservation area and non-eligible area.

3. Planting area is NOT eligible, if the land: a. was deforested to generate CO_2-certificates OR b. is wetland OR c. is situated on ground that is permafrost OR d. is agriculture farming land and threatens the local production of staple food through its conversion to forest.	
a.	See answer at point 2.a
b.	The plantation areas are located in the Andes at a height of 2700-3500m. No wetland is present there.
c.	The plantation areas are located in the Andes at a height of 2700-3500m. No permafrost is present there.
d.	The designated plantation areas have been appointed for reforestation by the local famers (stakeholders) in the project area. This was the result of extensive consultation and informative sessions with the local communities (land owners and - users) in the area. According to the communities, the assigned plantation areas are not being used and will not be used for agricultural or silvipasture activities.

4. Evidence must be given, that in case any agricultural or silvopasture activities are taking place on the project area, they contribute to the aim of creating a forest.
No agricultural or silvipasture activities are planned on the reforestation areas.

5. Evidence must be given that no project activities will lead to a long-term increase of emissions within the carbon pool 'soil'.

No area wide ploughing will be utilized before the planting activities. Instead, planting holes will be created (40x40x40cm) for each individual young tree. This means that the soil between the planted trees will remain largely untouched. Some of the vegetation will be removed around the planting hole (around 1m) if this vegetation causes too much competition (sunlight, water) with the young tree and/or to easy digging of the planting hole. The pruning material will be mixed with the bare soil around the young tree to prevent spontaneous/natural fires.

The photos below show how activities are carried out: People only use hand tools for land preparation.



Photo 6: Use of manual equipment to dig nurseries beds.



Photo 7: Use of shovel to dig plant holes.

03 Forest Management

Name of the Project	Reforestation Sierra Piura

1. A description of the project's silvicutural objectives must be given.
The silvicultural objective of the reforested areas are (quality) timber by selective harvesting practices. Besides this, the project must generate wood products for the caserios involved in the project. Wood products needed by the community are: firewood (for cooking and housewarming), construction wood and wood for making tools. A number of pruning and thinning activities will be carried out. The intensity and point of time of thinning and pruning depends on the tree species involved, but on average the first thinning will be carried out at 7-8 years after planting and the second thinning at the age of 14-15. Only *Alnus jorullensis* requires a quicker (but low intensity) thinning at age 3. However, the optimal thinning intensity and moment will be determined when the time is near. Pruning will be carried at 4-6 years after planting. This activity is only necessary for the Pinus species, since these species are poor in self-pruning. *Alnus jorullensis* is very well in self-pruning, but in case fire wood is needed, this species can also be pruned at the age of 5-7. The *Polylepis incana* stands will not be pruned nor thinned and will solely be used as conservation area. Except when the health and growth of *Polylepis incana* or its undergrowth threatens to stagnate appropriate management will be applied. The long term strategy (20-25 years) is maximizing timber volume production, in agreement and harmony with the environment and the caserios involved. This means that possible negative impacts will be prevented beforehand or forest management will be adjusted during the runtime of the project if it appears that certain activities have a negative impact on the environment. An additional and mutual important objective of the project is to reintroduce a forest ecosystem with a certain percentage of native tree species, such as *Alnus jorullensis* and *Polylepis incana*. However, other species might also be considered. Some research and inventories has already been carried out to examine native tree species that are suitable for reforestation in the project region, such as the research done by Cotler and Maass (1999) (ref: 03_06_Cotler_H_and_Maass_JM_1999). This results of this research can be used to carry out some testing and to inform/teach the farmers about the alternatives for Pinus species. It should be mentioned that in the first rotation of the plantation, the forest ecosystem will be that of a forest plantation and as such not will be comparable with an ecosystem of a natural forest. The aim after/during the first rotation is to convert the plantation forest gradually to a forest-like ecosystem with a higher diversity of native tree species and a higher diversity in horizontal en vertical structure. The plantations will continue to be used as a timber resource by the farmers but on a sustainable and balanced manner. As such, awareness raising and education in forest management among the local people will also be an important part of the project. The first rotation will take about 25 years. During this first rotation period several activities will be carried out, such as harvesting activities (pruning and thinning), fire control (management of fire corridors) and disease control. The extent of the harvesting activities (rate and time periods) depend on the tree species concerned and will be tuned accordingly. After this period, it is expected that most planted trees have been harvested and young trees have established again by natural regeneration or by active replanting. At the start and during the first rotation period several small trail plots will be established within the plantations in which mixtures with several native tree species will be tested. This knowledge will be used to support the conversion management and to teach the local communities about the management of

these new tree species. Such knowledge is important because from literature studies it appeared that practical knowledge regarding the introduction and establishment of native tree species (in similar environments of that of the project area) seems to be limited or absent, especially related to conversion management (mixtures of Pinus species with native species). The final harvest will be gradually carried out in a time period of 5 years (using various harvest cycles).

All mapping material that has been developed for the project can be found in reference: *03_10_All_map_material_Piura.zip.* The shapefile with all management units and additional attribute information (planting year, caserio, etc) can be found in reference 03_12.
A collection of photo material that gives an impression of the project area and activities can be found in reference: *03_08_Photo_material_project_Area.zip*.

2. Evidence must be given that the borders of the project area, planting area (eligible and non-eligible), management units and nature conservation area are clearly defined and visible in the field.

The outer limits of the plantation areas will be clearly indicated in the field. Two methodologies will be used for this:

1. Sticks/Poles
   The outer limits of the project area (= planting area), managements units as well as the conservation area will be delimited with sticks. The sticks will be made from woody species (bushes) found in the area. Photo 1 below shows an example of a fence made with the same material (Populated centre of Choco). This type of fence will be used for the outer limits of the project area and will protect the young trees against grazing. The number of sticks will be lower for the separate management units since these are purely necessary for delineation and not necessarily for protection against grazing.

2. Raising of earthen walls or mounds (see photo 2a and 2b below):
   This system will only be used on the outer limits of the project area where goats threaten the young trees.



Photo 1



Photo 2a

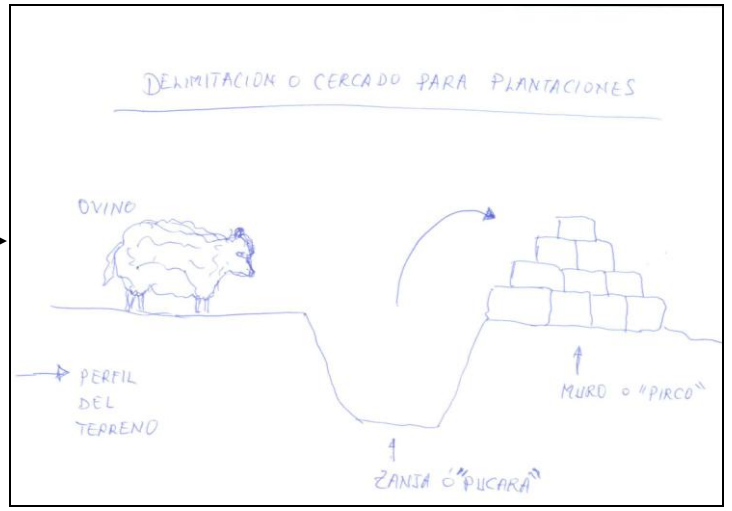


Photo 2b

3. Using firebreaks
   In between the seperate management units, firebreaks of 5-8 meters width will be created. In these firebreakes vegetation will be removed. The firebreaks will be maintained (trimmed and plowed) during the project cycly. When the trees are more mature, these firebreaks will be clearly visible in the field. As such, the firebreak acts as a delineation of the management unit.

Note: For plantation limits that coincide with exising delineations (footpaths, gorges, etc) no fences or earthen walls will be used. These parts are sufficiently protected already.

3. A description of the following tree species characteristics must be given:
   a. Origin and distribution of the tree species (indicate if the species are native or not)
   b. Provenance of the seeds
   c. Main purpose / use of trees
   d. Possible pests and diseases
   e. Time when forest products are foreseen to be used

	Name (English, Latin and if possible local language)	**Origin and distribution**	**Native?**	**Provenance**	**Main purpose / Use of trees**	**Possible pests and diseases**	**Time of use**
1	*Pinus patula* ('Pino')	Native to Mexico, where it grows on eastern slopes at elevations between 1650 and 3000m above sea level. Later introduced in Southern Africa, New Zealand, Australia, etc. (03_05_Lambrechts_1989)	No	Seeds will be purchased. The seeds must be certified and prove must be provided that the material is not genetically modified.	**General uses:** Crates, packing material, light cabinetwork, particleboard, paper pulp, pallets, furniture, etc. **Project uses:** firewood (branches), pulpwood, furniture, pallets and construction	**At nurseries:** *Rhizoctonia solani*, *Fusarium sp.* and *Pythium sp.* Rhizina undulate (root disease) Diplodia pini (if frequent hailstorms occur) Lepidiptera (leaf eating insects) Bole and tip break (in windy areas) Windthrow (in windy areas, but this is very rare) Fire (especially young trees and seedlings)	Firewood: after 4-6 years Thinning: after 14-16 years Quality timber: ≈ 25 years
2	*Pinus radiata* ('Pino')	Originally grows on narrow coastal strip in California and on the Island of Guadeloupe at elevations of 350m above sea level. Later on it has been planted in Australia, New Zealand, South Africa and Latin America (Chile).	No	Seeds will be purchased. The seeds must be certified and prove must be provided that the	**General uses:** Chemical pulp, indoor and outdoor construction, veneer, plywood. **Project uses:** Chemical pulp, indoor and outdoor	**At nurseries:** *Rhizoctonia solani*, *Fusarium sp.* and *Pythium sp.* No mayor threads have been observed. Rare threads that have been observed are:	Firewood: after 4-6 years Thinning: after 14-15 years Quality timber: ≈ 25 years

		(03_05_Lambrechts_1989)		material is not genetically modified.	construction, furniture wood, firewood	Rhyacionia buoliana (shoot borer, observed in Chile) Fungal diseases (in occurance of constant high atmospheric humidity): Fomes annosus, Armillaria melea, Diplodia pini and Cronartium cerebrum	
3	*Alnus jorullensis (=Alnus acuminate)* ('Aliso')	Native to Latin America, from Mexico to Central America and along the Andes from Venezuela, Colombia, Ecuador, Peru and Bolivia to Chile and Argentina (03_05_Lambrechts_1989)	Yes	Seeds will be purchased. The seeds must be certified and prove must be provided that the material is not genetically modified.	**General uses:** Construction, Crates, Household utensils, broomsticks, musical instruments, piles. Plywood, particle board construction, charcoal, albeit. **Project uses:** Chemical pulp, indoor and outdoor construction, furniture wood, firewood	The main factors limiting growth are underbrush, defoliating ants, and fungal attacks at the nursery stage. The most limiting factor for its establishment is low humidity, both in the soil and in the atmosphere. Water stress causes a browning of the leaf veins and then yellowing of the leaf. Accoding to Lambrechts (1989), A. jorullensis is not affected by any major pests or diseases. Only the fungi *Armillaria puriggarii* and *Pholiota aurivella* cause damage in some localities. Damage by cattle (tramping) of the young trees should be prevented.	Firewood: after 2-3 years Thinning: at 3-7-15 years Quality timber: ≈ 25 years
4	*Polylepis incana*	Ecuador, Peru and	Native in Peru.	Seeds will	**General uses:**	Grazing and trampling	> 30 years

	(‘Quinual’)	Colombia	Uncertain if it is native in the project area.	be purchased. The seeds must be certified	Firewood, construction wood, medicines	by herbivores! Caterpillars (Geometridae).	(trees grow very slowly!) Species not considered for intense exploitation

4. Evidence must be given that at least 10% of the project area is managed as a nature conservation area.

A total of 23ha of the native tree species *Polylepis incana* will be planted to comply with the minimum of 10% conservation area. This is about 10% of the total plantation area, which is 213ha. The *Polylepis incana* plantations will be extensively managed and will not be included in the final harvest.

In Peru 15 Polylepis species are growing, which makes Peru the most diverse country with regards to this species. Nevertheless, detailed studies of Polylepis forests in Bolivia and Peru have shown that this genus only occupies a fraction of the potential cover; around 2-3% for Peru and around 10% for Bolivia (Ref: 03_02_Ramsay&Aucca, page 1). Paleontological evidence and other studies suggest that Peru once was occupied by 55.000 km2 of Polylepis forest, but these superficies decreased by human overexploitation of these forests (between 700 and 1200km2 of forest was cut). Polylepis forests are an now a threatened ecosystem that is disappearing fast and as such these forest types need to be studied and protected urgently.



Photo 1



Photo 2



Photo 3

Photo 1 – 3: Photos of remnants of *Polylepis incana*, taken at caserio of Choco. See ref: *03_11_A4_Conservation_areas.pdf* for the locations that will be

reforested with *Polylepis incana*.

5. Evidence must be given that the nature conservation area is protected or managed in order to establish or re-establish the natural ecosystem of this area.

The nature conservation areas of this project will consist of *Polylepis incana* plantations that will be managed with the objective to preserve this area for future generations. As such, the *Polylepis incana* stands will not be harvested for construction wood after the first rotation (after 25 years). The practice however learns that some firewood harvest from forest resources is sometimes inevitable. However, in agreement with the local farmers, the objective is to preserve the Polylepis stands and to create a natural forest ecosystem that will attract specific mammals and birds and contribute to the biodiversity in the area.

In the conservation areas, *Polylepis incana* will be planted in a 3x3m spacing. The plants will be planted in holes of 40x40x40cm. Preferably, the plant holes will be filled with moistures soil that is treated with cheap dung or fertilizer (soils often lack Phosphorus). Expected growth will be in the order of 20-65 cm/year in height and about 3.4 to 17.6 mm/year in diameter (with proper treatment). Of course this largely depends on the site conditions (altitude and soil). Experiences have proofed that survival rates of Polylepis at 3800-4200 masl are very high (90% - 100%). (Ref: 03-04_polylepis incana_Aquirre_Martinez, page 255 § 7, page 260 table 2, page 261 §5, page 264 §4 and page 265 §3).

After the planting, the management of the area will be extensive. The main objective is however to save guard the forest health and growth of these stands. Management practices will be carried out according to this objective. Only minor corrective actions will be carried out with respect to forest management if this is needed. According to Rohner C. (Ref: 03_03_ Christian_Rohner), Polylepis plantations will not provide quality timber without proper management. If Polylepis stands are not managed, the timber will not be suitable for construction purposes and will only serve as firewood. However, according to Rohner, this is exactly how Polylepis stands naturally should grow: Native and naturally grown Polylepis forests tend to put their energy more in the competition for soil cover (soil nutrients) than competing for height (light). Due to these characteristics Polylepis spp. are excellent to combat soil erosion. Another advantage of extensive/no management is that the tree will branch off close to the ground. These branches will form tree roots as soon as they come into contact with the soil, creating new tree juveniles again. Hence, according to Rohner, Polylepis plantations preferably need minimal interventions to create a native ecosystem that protects the soil and may supply some firewood in the future.

6. Key figures on the following parameters must be provided:
   a. Project area
   b. Foreseen planting area
   c. Eligible area within the foreseen planting area
   d. Nature conservation area

   In case the areas are fragmented, a table shall give a clear overview.

a.	Project area	253 ha
b.	Foreseen planting area	213 ha
c.	Eligible	84 %
d.	Nature conservation area	10 ha

Note: The conservation area (10ha) is being planted with *Polylepis incana* and is part of the total planting area (213ha). Hence, the total area that will commercially be exploited is 213 ha - 10ha = 103ha.

In case the project area is fragmented, please give each area an ID and use the table below. Otherwise delete this part.

ID Management units	Name area	Project area (ha)	Eligible Planting area (ha)	Eligible (%)	Nature conservation area (ha)
AH1	Alto Huancabamba	3	3	100%	0
AH3	Alto Huancabamba	1	1	100%	0
AM2	Alto Mayo	1	1	100%	0
AM3	Alto Mayo	1	1	100%	0
CA2	Cajas	2	2	100%	0
CA7	Cajas	2	2	100%	0
CH1	Choco	3	3	100%	0
CH2	Choco	5	5	100%	0
CO4	Confesionarios	2	2	100%	0
CO5	Confesionarios	1	1	100%	0
LH1	Las Huacas	1	1	100%	0
AH2	Alto Huancabamba	1	1	100%	0
AH4	Alto Huancabamba	1	1	100%	0
AH5	Alto Huancabamba	4	4	100%	0
AH6	Alto Huancabamba	1	1	100%	0
AM4	Alto Mayo	3	3	100%	0
AM5	Alto Mayo	1	1	100%	0

AM6	Alto Mayo	2	2	100%	0
CA3	Cajas	4	4	100%	0
CH3	Choco	5	5	100%	0
CO1	Confesionarios	2	2	100%	0
CO3	Confesionarios	1	1	100%	0
CT1	Chontali	7	7	100%	0
LH2	Las Huacas	3	3	100%	0
LH4	Las Huacas	1	1	100%	0
SC1	Santa Cruz	3	3	100%	0
SC5	Santa Cruz	2	2	100%	0
SL1	Sargento Lorenz	3	3	100%	0
SL2	Sargento Lorenz	4	4	100%	0
AM1	Alto Mayo	1	1	100%	0
AM7	Alto Mayo	1	1	100%	0
CA5	Cajas	6	6	100%	0
CA6	Cajas	1	1	100%	0
CA9	Cajas	3	3	100%	0
CO2	Confesionarios	1	1	100%	0
CO6	Confesionarios	3	3	100%	0
CO7	Confesionarios	1	1	100%	0
CT10	Chontali	4	4	100%	0
CT7	Chontali	4	4	100%	0
CT9	Chontali	3	3	100%	0
HU1	Huambiche	2	2	100%	0
HU2	Huambiche	2	2	100%	0
LH3	Las Huacas	3	3	100%	0
SC3	Santa Cruz	4	4	100%	0
SC4	Santa Cruz	2	2	100%	0

CT11	Chontali	2	2	100%	0
CT13	Chontali	1	1	100%	0
CT3	Chontali	1	1	100%	1
CT4	Chontali	7	7	100%	0
HU3	Huambiche	2	2	100%	0
LH5	Las Huacas	3	3	100%	0
SC2	Santa Cruz	1	1	100%	0
SL3	Sargento Lorenz	1	1	100%	0
SL4	Sargento Lorenz	8	8	100%	0
CA1	Cajas	2	2	100%	0
CA10	Cajas	3	3	100%	3
CA4	Cajas	14	14	100%	0
CA8	Cajas	1	1	100%	0
CT12	Chontali	2	2	100%	0
CT14	Chontali	2	2	100%	0
CT15	Chontali	14	14	100%	0
CT2	Chontali	20	20	100%	0
CT5	Chontali	4	4	100%	4
CT6	Chontali	2	2	100%	2
CT8	Chontali	2	2	100%	0
HU4	Huambiche	3	3	100%	0
SC6	Santa Cruz	2	2	100%	0
SL5	Sargento Lorenz	5	5	100%	0
Non Eligible 1	Cajas	4	0	0	0
Non Eligible 2	Cajas	9	0	0	0
Non Eligible 3	Choco	4	0	0	0
Non Eligible 4	Confesionarios	2	0	0	0
Non Eligible 5	Alto Huancabamba	4	0	0	0

Non Eligible 6	Las Huacas	1	0	0	0
Non Eligible 7	Choco	4	0	0	0
Non Eligible 8	Chontali	12	0	0	0
TOTALS		**253**	**213**		**10**

04 Environmental Aspects

Name of the Project	Reforestation Sierra Piura

1. A description, including pictures, of the different land-use classes (see chapter 'Transparency' - Point 3c.) of the project area must be given.

   In case significantly different land-use classes are bordering the project area, they must also be described.

The foreseen plantation areas are characterized by 2 types of vegetation classes. One vegetation class is dominated by grassy vegetation such as: Ichu, Silcique and Pajilla Andina (see photo 1 below)



Photo 1: Landuse class with predominantly grass species.

A second vegetation class found in the plantation areas is also dominated by pasture vegetation, but this vegetation type shows a more patchy structure with mixtures of grasses and some woody bushes. This vegetation is about 30-50cm average in height. Species that were found in this vegetation class were: Huacun, Sarcilleja, Ichu, Silcique, Chilco, Sarcilleja, Hierba dulce, Paja, Pega –Pega and Cargara. Photo 2 below shows an example of this vegetation class:



Photo 2: Patchy pattern of vegetation and a mixture of grass species and woody plants

Other species that are recorded in the surrounding area of the plantation areas are: *Stipa brachyphylla*, *Stipa mucronata*, *Festuca parviniculata*, *Paspalum tuberosum*, *Calamagrostis antonima*, *Agrostis tolucencis* and *Disticha muscoides*, *Baccharis latifolia*, *Chusquea spp*., *Lepechinia spp*. and *Clusia spp*. The natural forest has almost completely disappeared due to firewood demand. A few small forest remnants have remained: It assumed that the closely neighbouring cloud forest remnant 'Mijal' (about 1.5 km North-West of the project area) is an example of the past vegetation cover in the project area. Species found in these remnants are: *Alnus jorullensis*, *Polylepis spp*., *Oreocallis grandiflora*, *Myrcia spp*. *Escallonia resinosa* and the native Ceder.
It must be emphasized that in the foreseen plantation areas no agricultural activities take place. Only a few sheep are hold, but these animals can only survive in the harsh environment due to their tenacity, not because of the nutritional value of the vegetation (which is very low). This is indicated by the weight of the sheep, which is about 6-10kg average. The conclusion is also supported by the fact that

1. A description, including pictures, of the different land-use classes (see chapter 'Transparency' - Point 3c.) of the project area must be given.

   In case significantly different land-use classes are bordering the project area, they must also be described.

the farmers clearly indicated the allowable reforestation areas and were determined not to appoint any other area that was of importance for agriculture or cattle farming: Suitable and productive areas in Choco are that scarce so that any soil with minor significant productivity is used by the farmers for their food supply.

Other landuse classes bordering the project area: The vegetation neighbouring the project area is similar of those described above, hence no distinct vegetation types have been found in the direct neighbourhood of the plantation areas. The videos in ref: 01_01_VIDEOS ELEGIBILIDAD also give an impression of the vegetation.
In the plantation area and its surrounding, no agricultural activities are carried out: The project area and direct surroundings are characterized by extensive pastures. On these pastures, a minimal number of sheep are held that can survive in these harsh circumstances (cold/wet climate, poor nutrition) due to their persistency. In the lower parts, at some distance from the project area, the farmers grow their crops, exclusively for subsistence farming.

2. Evidence must be given that the project has net-positive ecological impacts.

   Therefore,
   a. positive ecological impacts of the projects must be enhanced and
   b. negative or potential negative ecological impacts must be mitigated.

   Descriptions of the following project characteristics must be given for both points:
   - Soil Nutrients, Erosion
   - Water Quality, Quantity
   - Biodiversity Flora, Fauna
   - Climate Temperatures, Rain

Soil	**Current situation** **Soil nutrients** The soils in the project area are of volcanic origin. The Andean soils found in Piura (and South Ecuador) are distinct and have less favorable properties than the Andean soils found in the North of Ecuador: They have lower moisture content and available water content, lower bulk density, less OM and N content, lower P retention capacity, and a coarse texture. They are connected with very recent and coarser ashes, lower precipitation, and (previous) inappropriate land use (ref: 04_04_Hofstede et al_2002, page 165, §2). Soil colors in the project area vary from dark to brownish-yellow to brownish-gray. Soil texture is defined as fine or loam-clay with superficial and deeper horizonts of fine and coarse gravels/rocks (see ref: 04_01_Soil_samples). The soils are low in available potassium (K) and phosphorus (P) content and as such are classified as poor to moderate in fertility. Due to their (relative) high organic matter content, the soils have a low pH value. **Soil erosion** Currently, no trees are found in or close to the planting areas. Most parts of the reforestation area are dominated by pasture of about 30-50 cm high. These pastures stabilize the soil and to some extent prevents erosion on the steep slopes. Slope degrees in the plantation areas are high and vary from 0% to 75%. Some slopes even exceed 75% (see ref: 04_02_Slope_Percentages). This makes the soils prone to erosion during heavy rains, which is actually occurring in a moderate scale in parts of the project area (see ref: 04_09_Erosion_during_heavy_rains). At lower elevation levels in the Piura basin the erosion is much severe, especially in the parts with agricultural activity. Here, the soil is loose and liable to erosion, especially on

the steeper slopes.

The current pasture vegetation does not buffer soil erosion processes entirely, especially in periods with precipitation peaks erosion gullies can develop, like in case of an El Niño event. In recent decades El Niño disturbances have resulted in costly damage in the watershed of the Piura River by causing landslides in steep areas and severe floods downstream. In 1998, the losses were valued at more than US$100 million. Rainfall in a 1983 event was even higher, but because of the high deforestation rate after 1983 the damage was greater in 1998 (Ref: 04_03_Barrena_M._etal_2007, page 1, §1).

According to Fernandez Barrena (Ref: 04_03_Barrena_M._etal_2007, page 2, §1) all the banks of the Piura River from the outlets of its main tributaries to its arrival at the Pacific Ocean are vulnerable to sudden rises in precipitation (e.g. caused by an El Niño event). Practically the whole zone, including most upland farmland (the project area forms part of these areas), has a high to very high erosion risk. Geographic information system (GIS) analysis showed that erosion risk is influenced more by forest cover than by soil type. The solutions for the erosion problem are clear: conservation of the remaining forests, recovery of lost areas, reforestation of stands, a shift to agroforestry and other systems that protect soil from erosion, and development of silvipasture (Ref: 04_03_Barrena_M._etal_2007, page 2, §1)

Positive aspects of project

Soil nutrients

There is some international discussion about the introduction of (large scale) pinus in the Andes and its potential negative effect on the eco-system, soil and hydrology. However, according to Hofstede et al. (2002), the effect of Pinus plantations on the ecosystems of the high Andean zone of Ecuador cannot be generalized (ref: 04_04_Hofstede et al_2002, page 165, §2). These effects vary depending on the region (soils and climate), previous land use, and present management of the plantation. However, in general Pine plantations in comparison with natural ecosystems (e.g. natural páramo grasslands) rarely have a positive effect on soils. Direct effects of soil degradation under Pinus plantations were on the other hand neither observed by Hofstede. Hence, according to Hofstede either positive or negative influences of the reforestation activities on soil nutrients can be expected. In the areas where *Alnus jorullensis* (=*Alnus. acuminate*) and *Polylepis incana* will be planted, the chemical characteristics of the soil will most probably improve. *Alnus jorullensis* is well known for its capacity to improve soil properties because the species establishes a symbioses with nitrogen fixing actinobacteria. This bacteria converts nitrogen into soil-soluble nitrates which favorable enhances soil fertility. As such, *Alnus spp*. benefits other plants growing near them by taking nitrogen out of the air and depositing it into the soil in usable form. In a 2-year-old A. *acuminata* (= *A. jorullensis*) plantation in the Colombian highlands (1200 trees/ha), Carlson and Dawson (1985) estimate an annual increase in soil nitrogen of 279 kg/ha (see ref: 04_20_Alnus_soil_properties, chapter: Symbiosis).

Soil erosion

Because tree roots penetrate the soil deeper than grass roots, trees will better stabilize the soil. Also, the physical characteristics of the soil might improve or remain stable. Pine plantations also tend to have higher richness in species understory composition. According to Hofstede, vegetation understory is sometimes positively influenced (increase in woody species), sometimes negatively influenced (no understory) or is neutral (ref: 04_04_Hofstede et al_2002, page 165, §2). However, Hofstede found a positive relation between the richness of understory vegetation and the openness of the tree canopy. This means that with proper plantation management (e.g. by sufficient, on time thinning and opening of the tree canopy), understory vegetation development can be positively influenced especially in the project area where woody species are already present. During the first 5-7 years after establishment the tree canopy will be closed, but according to Hofstede

this will not negatively influence the present woody vegetation cover. The increase in woody vegetation cover will decrease soil erosion.

Also a reduction in grazing animals will reduce soil erosion. According to Hofstede (ref:04_04_Hofstede et al_2002, page 165, §3), a result of trampling is that soils are denser, drier, less organic, and less acid. Grazing and trampling also prevents the regeneration of woody species, and the understory is transformed from tussock grassland into short, pasture-like grassland. These effects are negative for soil erosion. Although the number of grazing animals in the plantation areas is currently small, lowering the number of grazing animals in the plantation areas to a minimum (even to zero) will reduce soil erosion.

Potential negative aspects of project

Soil Nutrients

The reforestation activities with Pine may negatively influence available soil water, soil moisture and organic matter content. Hofstede (ref: 04_04_Hofstede et al_2002, page 165, §2) found lower values for the complex of interrelated factors (moisture content, available water, OM content, P retention, and texture) in pine plantations compared to those in parámo vegetation. However, he also mentions that this analysis was only partly confirmed by direct statistics. The effects vary depending on the region (soils and climate), previous land use, and present management of the plantation (ref: 04_04_Hofstede et al_2002, page 165, §2). During informal interviews with farmers that live in or close to the project area, these farmers indicated that they haven't experienced negative impacts of OM content and neither a decreasing soil water content. In the contrary: They even experienced increases in soil water content. This conclusion is supported by two Peruvian forestry experts who mentioned that soil water content in Pinus plantations is only a problem if precipitation is lower than 300mm/year (see ref: 04_18_Plantaciones forestales). In their practical experience they observed an increase in soil water content in various large scale plantations in the Peruvian Andes. This supports the conclusion that the effects of high altitude pinus plantations on Andean soils are very much depending on the circumstances in the area (precipitation, soils, current vegetation) and that there is still some discussion on this topic.

Soil Erosion

According to Buytaert et al., (2007) (ref: 04_05_Buytaert_etal_2007) the high infiltration and water storage capacity of the páramo soils are closely related to their exceptionally high organic matter content. *P. patula* and *P. radiata* species have a considerable water consumption and a strong effect on the evapotranspiration in the higher parts of the Andes that was formerly grown by the natural vegetation, such as páramo. Afforestation activities with *P. patula* and *P. radiata*, can reduce soil organic matter contents because of a faster OM decomposition due to the lower soil water content and because organic matter production in the soil is lower under pinus plantations. Soil erosion already occurs in the project area on a moderate scale (see ref: 04_09_Erosion_during_heavy_rains) and in the lower parts of the Piura basin at a very large scale (see 04_15_CONAM_climate_change_and_adaptations). This indicates that currently, both soil organic matter content and the pasture vegetation in the higher parts of the Piura basin are not sufficient to buffer the amount of water runoff. Hence, in this case reforestation activities would most certainly help to reduce erosion. In any case it is not expected that the reforestation activities will negatively influence soil erosion.

Mitigating activities of negative aspects

Soil Nutrients

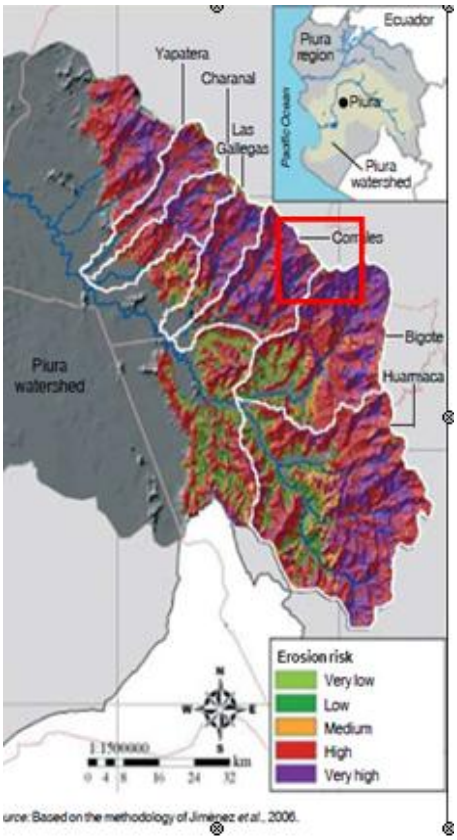
Hofstede (ref: 04_04_Hofstede et al_2002, page 166, §2) recommends to use integrated forestry techniques, including native tree species, which probably reduce possible negative impacts on the Andean ecosystems and may form a sustainable alternative to massive plantations with exotics in the future. This is exactly what the

	project in Piura is aiming for: A mix of exotic pinus species with a considerable percentage of native species (*Polylepis incana & Alnus jorullensis).* The proposed mix of tree species also meet the economic demands of the farmers. The silvicultural objective is a gradual forest conversion to a higher percentage of native species after the first rotation (>= 25 years). This means that the pinus plantations will gradually be converted to a more natural-like forest with a higher percentage of native tree species. By doing so, potential negative effects that are mentioned by Hofstede and Bytaert will be mitigated. Another mitigating activity is being created by using native species (*Polylepis incana & Alnus jorullensis*). Both species are planted in considerable percentages and will contribute to the soil nutrient balance. Especially *Alnus jorullensis* has great potentials due to its nitrogen fixing capacity (see ref: 04_20_Alnus_soil_properties, chapter: Symbiosis). **Soil Erosion** The results brought up by Buytaert et al., (2007) (ref: 04_05_Buytaert_etal_2007) concerning the impact of reforestation on soil water content is certainly something to consider. By reforesting the area, the soil water content (and thus the soil organic matter content) may decrease, because all proposed tree species potentially consume more soil water that the pasture species that are now growing in the plantation areas. However, this effect is still uncertain and according to Hofstede (ref: 04_04_Hofstede et al_2002, page 165, §2) this largely depends on the location (soil and climate). Moreover, farmers and forestry experts indicated from practical experience that soil water availability in their areas increased after reforestation with Pinus (informal interviews held with farmers & ref: 04_18_Plantaciones forestales). The plantation areas proposed for this project are already degraded and the soil organic matter content, nor the vegetation, are comparable with values/situations found in a natural páramo. Soil samples that were taken in the plantation areas indicate that soil organic contents are not very high, not even in the top layer (horizont). Hence soil organic matter content and soil water quantities may already have been declined during the time that the area was deforested and converted to pasture. This is also concluded by the huge amounts of erosion in the lower parts of the Piura basin (see ref: 04_15_CONAM_climate_change_and_adaptations). It may thus be possible that the soil water content will actually be altered by the reforestation activities, since the a combined pasture-tree vegetation (the tree species and the wide tree spacing do not hamper the growth of grasses and plants) will catch more air moisture than solely the pasture vegetation alone. Furthermore, the scale of reforestation is relatively small and fragmented and the used tree density will not be high (3x3 meter). Sufficient thinning will be carried out during the first rotation, which will open the tree canopy and allows understory growth of woody species and plants (ref: 04_04_Hofstede et al_2002, page 165, §2). Hence, by proper management of the plantations, undergrowth will be stimulated which also reduces soil erosion.
Water	**Current situation** **Water quality** The plantation areas are located on mountain ridges where no large water bodies have been found, except six small water pools (see ref: 04_08_Waterpools_in_projectarea). The map 'Hydrology' (see ref: 04_06_Hydrology) shows the locations of rivers/gullies and water pools in the lower parts in the surrounding of the project area. The rivers were mapped by using a topographic map, scale 1:25.000 of (Ministry of Agriculture, COFOPRI). The project area is currently vegetated with (degraded) pastures and some woody species that grow in low densities. Issues related to water quality have not been found or reported by the farmers. However, the mud streams along the unpaved roads during heavy rainfalls may affect water quality at lower elevations to a certain extent and may silt up water bodies (e.g. the water pools) with soil sediments (ref:

04_09_Erosion_during_heavy_rains). The scale of this effect is however unknown and as said, not reported as a major problem by the farmers in the project area. This is also due to the fact that the project area is located in the (most) upper part of the Piura basin where the water base flow is relatively low compared to the water flows in the lower parts of the basin. However it is assumed that water quality will almost certainly deteriorate during an event of El Niño.

Water quantity

No problems have been reported by the farmers related to shortages in potable water nor in water excesses. An explanation for this may be that farmers became accustomed with the situation and lost reference with the past situation. Excesses in water-runoff seems to be the biggest problem, although even this seems to be a moderate problem at the project location. However, water runoff can potentially become a bigger problem in case of an Niño event or in case the vegetation in the area further degrades. The photos in the appendices (see ref: 04_09_Erosion_during_heavy_rains*)* show the current erosion effects along unpaved roads during heavy rainfall. In the lower areas of the Piura Basin, erosion due to excesses in water runoff can be a tremendous problem during an el Niño event. Barrena et.al. (2007) (ref: 04_11_Barrena_etal_2007, page 51) mentions that the project area is located in an area that is prone to strong erosion (see map below).



Map 1: Erosion risk in Piura Basin (project area is located in the red square) by Barrena et al. (2007)

According to Barrena (see ref: 04_11_Barrena_etal_2007, page 51, §4) The project area belongs to one of the six main subwatersheds (see map 1 above) where improvements could contribute to regulating the hydrological cycle. All of the banks of the Piura River from the outlets of its main tributaries to its arrival at the Pacific Ocean are vulnerable to sudden rises. Practically the whole zone, including most upland farmland, has a high to very high erosion risk (see map 1 above). Geographic information system (GIS) analysis showed that erosion risk is influenced more by

forest cover than by soil type. According to Barrena, the solutions to the problem are clear: conservation of the remaining forests, recovery of lost areas, reforestation of stands, a shift to agroforestry and other systems that protect soil from erosion, and development of silvipasture.

To a certain extent, the current pastures lowers the quantities of water-runoff compared to a bare soil situation. However, according to Barrena (see ref: 04_11_Barrena_etal_2007, page 51, §4) this vegetation cover does not provide enough buffer against erosion. Especially for the areas in the lower basin, where erosion is most severe, the buffering effect provided by pastures in the upper areas is not sufficient.

Positive aspects of project

Water quality

The project will contribute to a lower water runoff from the upper areas into the areas located in the lower basin. As such, silting up of the water bodies located in the lower areas will decrease and water quality will increase.

Water quantity

Water quantities in the form of water-runoff will almost certainly decrease because of the reforestation activities. Farley, et al. (2005) (see ref: 04_07_Farley_KA_etal_2005) examined the differences in the yields of water runoff for grass lands, shrub lands and afforested lands. They found that annual runoff was reduced on average by 44% (± 3%) and 31% (± 2%) when grasslands and shrublands were afforested, respectively. Eucalypts had a larger impact than other tree species in afforested grasslands, reducing runoff by 75% (±10%), compared with a 40% (± 3%) average decrease with pines. According to Farley, the results of this investigation clearly demonstrate that reductions in runoff can be expected after afforestation of grasslands and shrublands and may be most severe in drier regions.

The investigation of Farley supports the assumption that the reforestation activities will reduce water runoff and erosion. This buffering effect may become increasingly important since the climatic prognosis for the project area is a yearly increase in precipitation due to climate change (and the el Niño events).

The reforestation activities may also have a positive impact on soil water content. According to informal interviews held in a nearby village (called 'Huacas') the farmers experienced positive changes in soils water content after reforestation with pinus. For example, they reported that the wells in their area are now fuller than before the reforestation activities.

Potential negative aspects of project

Water quality

No negative effects on water quality are expected due to the reforestation activities. Albarino mentioned that reforestation of riparian areas by tree monocultures may affect the detrital foodweb of the river streams, which may have consequences for the water quality (and quantity) (ref: 04_10_Albarino_R._date_unknown, page1, §2). However, the plantation areas are not located close to any water body with a vulnerable and complex ecosystem that can be affected. As can be seen on the photos (see ref: 04_08_Waterpools_in_projectarea) the water pools in the project area are not significant from an ecological point of view. As such, no negative effects are expected of the reforestation activities on water quality.

Water quantity

According to Buytaert, W., Iñiguez, V. and De Bièvre, B., (2007) (see ref: 04_05_Buytaert_etal_2007, page 2, §1) it is generally accepted that total water consumption of forests is larger than the water consumption of short vegetation such as shrublands and grasslands. Forests generally have a higher leaf area index and

	roughness, and a deeper and better-developed root system, resulting in a higher transpiration. Interception and subsequent evaporation from the canopy and the litter layer also tends to be higher in forests. However, the magnitude of the change in water yield due to afforestation can vary considerably, depending on climatic conditions, the type of original and replaced vegetation, its age, and the type of activity. Factors such as the total forested area and tree density have an impact as well and are not always easy to determine (see ref: 04_05_Buytaert_etal_2007, page 2, §1). Farley et al. (2005) (see ref: 04_07_Farley_KA_etal_2005, page 1) summarize the 26 case studies of afforestation, 13 of which were originally grassland. The studies reveal that in general water runoff reduces by 44% (± 3%) on average and 31% (± 2%) when grasslands and shrublands are afforested, respectively. Eucalypts had a larger impact than other tree species in afforested grasslands, reducing runoff by 75% (± 10%), compared with a 40% (± 3%) average decrease with pines. Runoff losses increased significantly with plantation age for at least 20 years after Planting. A plantation age of 26–30 years coincides with the maximum observed decrease (see ref: 04_07_Farley_KA_etal_2005, page 1). The study of Farley and Buytaert reveal that overland water flow regimes can be both positively or negatively affected by reforestation activities. This depends on the rainfall intensities and the current vegetation in the area. This was also concluded after consultation of a Peruvian forestry experts that has gained a lot of experience with Pinus plantations in the Andes. He mentioned that potential negative effects of Pinus plantations on soil water quantities will only occur at elevations lower than 1500m where precipitations in general are lower than 300mm/year (see ref: 04_18_Plantaciones forestales). Current rainfall intensities already has caused huge erosion problems in the lower areas of the Piura basin (see ref: 04_11_Barrena_etal_2007). This indicates that the current vegetation in the upper basin areas does not sufficiently buffer overland water flows. Since it is foreseen that precipitation will increase in the future due to climate change, it is important to consider reforestation activities the upper basin areas to reduce these overland water flows (also see ref: 04_11_Barrena_etal_2007, page 1). Nevertheless, the articles of Farley and Buytaert bring to the attention that intensive reforestation of pasture areas by large tree monocultures (both conifers and broadleaf trees), should be carefully considered and wisely applied (ref: 04_07_Farley_KA_etal_2005, page 1566, §1). The project developers have taken these considerations seriously by planting more than 20% of the area with native (broadleaf) trees species (*Polylepis incana* and *Alnus jorullensis*) taking into account also the farmers' preferences for Pinus species. **Mitigating activities of negative aspects** **Water quality** No negative aspects are to be expected on water quality. **Water quantity** The project aims to reforest the areas with a mixture of tree species, both conifers (*Pinus spp.*) and broadleaf trees (*Alnus jorullensis* and *Polylepis incana*). The pinus spp. have been selected because their low demands in growing requirements and for socio-economic reasons: The communities have expressed high preference for pinus spp. (*P. radiata* and *P. patula*) and the farmers already have a lot of practical experience with both species. A considerable percentage of the plantation areas will also be plantated with the broadleafed native species *Alnus jorullensis* and a smaller percentage with the native *Polylepis incana*. The project has aimed to create a proper balanced mix of tree species so that the project is economically viability,

ecologically justifiable and entirely adopted by the community. Hence, a tradeoff has been made between poverty alleviation, economic- and ecological risks.

According to Farley, reforestation of grasslands and shrublands causes a decrease in water runoff, regardless the tree species that are used. Especially in areas where water runoff is less that 10% of the mean annual precipitation, one should be cautious with reforestation and the ground water balance.
However, with respect to the research carried out by Farley et.al. and Buytaert et.al. (see Ref: 04_05 and 04_07) we can already contradict some of these outcomes in relation to the project activities in the Sierra Piura due to practical experiences of the farmers with forest plantations. During interviews held with farmers of a nearby village called Huacas, the people indicated that they experienced increases in soils water content after reforestation their areas with Pinus some 8 years ago: The wells in their project area were fuller than before the reforestation. This observance contradicts the research results of Farley, who states that in all stages of the growing cycle, water runoff tends to decrease.

To mitigate potential negative effects on water quantities in the project area, the reforestation activities will be carried out on a fairly small scale and tree densities will be conservative (3mx3m or 2.6mx2.6m). Tree densities will decrease even more after the first thinning, that will be carried out around 5-7 year after planting.

Biodiversity

Current situation

Flora

According to dated classifications of INRENA (National Institute of Natural Resources of Peru), the vegetation in the project area is classified as 'Monte Subhumedo semicalido', 'Monte Humedo Templado' and 'Pajonal Altoandino'.

To determine the wet-dry ratio of the biomass, twelve sample plots (1x1m) in the area were taken. For these sample plots also the flora was determined. The following species have been found:

Pteridofita (Helecho)

Local name	Scientific name	Family
"Jargara"	Pteridium aquilinum (L.) Kuhn.	Dennstaedtiaceae

Monocotiledonea (Liliopsida):

Local name	Scientific name	Family
"Ichu"	Stipa ichu (Ruiz & Pavón) Kunth	Poaceae

Dicotiledonea (Magnoliopsida):

Local name	Scientific name	Family
“Lanche”	Myrcianthes discolor (H.B.K) Mc Vaugh	Myrtaceae
“Hierba dulce”	Muehlenbeckia volcanica (Ben.) Endl.	Poligonaceae
“Panza de vaca”	Lepechinia sp.	Poligonaceae
“Chilca”	Baccharis latifolia (R&P) Persoon.	Asteraceae
“Chilco”	Baccharis glutinosa Pers.	Asteraceae
“Huacun”	Baccharis grandicapitulata Hieronym.	Asteraceae
“Ushpa”	Chuquiraga sp	Asteraceae
“Huamingo”	Berberis lutea R&P	Berberidaceae
“Poleo”	Mentha pulegium L.	Lamiaceae
“Sarcilleja”	Brachyotum quinquenerve (Ruiz & Pav.) Triana	Melastomaceae
“Garado”	Calceolaria sp	Scrophulariaceae

Fauna

To investigate the fauna species that are currently present in the area, the communities were asked which birds and mammals have been observed by them. First, a list of potential species was established from an flora and fauna investigation in a neighboring páramo area called Pacaipampa, situated about 11 km north of Cajas. Although the vegetation in the project area is not entirely similar with that of Pacaipampa, it is expected that a few birds and mammals that were found in that Pacaipampa, could also be found in the plantation areas. The following species were found in Pacaipampa:

Birds found in Pacaipampa (see ref: 04_12_Garcia_D.etall_2007_DO NOT PUBLISH THIS DOCUMENT.doc, Chapter IV EVALUACIÓN DE LA AVIFAUNA, chapter 2: Pacaipampa):
Anas andiu, Falco femoralis, Phalcobaenus megalopterus, Larus serranus, Caprimulgus longirostris, Lafresnaya lafresnayi, Metallura odomae, Chalcostigma herrani, Ptherofanes cyanopterus, Eriocnemis vestitus, Cinclodes fuscus, Margarornis squamiger, Schizoeaca griseomurina, Leptasthenura andicola, Asthenes fammulata, Colaptes rupicola, Scytalopus canus, Mecorerculus leucophrys, Octhoeca fumicolor, Turdus fuscater, Notiochelidon murina, Antus bogotensis, Troglodytes solstitialis, Diglossa humeralis, Diglossa laresnayi, Anisognaths igniventris, Thlypopsis ornata, Phrygilus unicolor.

Mamals found in Pacaipampa (see ref: 04_12_Garcia_D.etall_2007):
Didelphis albiventris, Tamandua mexicana, Dasypus novemcictus, Lycalopex culpaeus, Tremarctos ornatus, Conepatus semistriatus, Eira barbara, Leopardus pardalis, Tapirus pinchaque, Odocoileus virginianus, Mazama americana, Scirus stramineus, Thomasomys sp 2, Akodon molllis, Agouti taczanowskii

Positive aspects of project

Flora

An established forest climate with an increasing mixture of tree species will probably alter the biodiversity in the area since (plantation)forests offer a better shelter for birds and mammals and a larger diversity in forage possibilities than the current pastures do. The tree density will be low 5-7 years after establishment, which enables vegetation undergrowth of grasses and favors the establishment of woody species. Hofstede (see ref: 04_04_Hofstede et al_2002, page 165, chapter Discussions & Conclusions, §5) found a positive relation between the richness of understory vegetation and the openness of the tree canopy.

Grazing animals will be kept out the forest plantations. Although grazing animals in the project area are currently low in numbers, this measure will certainly be beneficial for certain flora species.

The flora diversity will almost certainly increase during the life time of the project due to the above mentioned measurements. With proper management, the tree canopy will be open and the vertical tree structure will diversify. All this allows a higher total richness in flora species, especially woody species (see ref: 04_04_Hofstede et al_2002, page 165, chapter Discussions & Conclusions, §5). Moreover, after the first rotation (25 years), the objective is to introduce a higher percentage of native tree species (e.g. *Polylepis incana* and *Alnus jorullensis*) by carrying out a conversion management. The gradual alteration in vertical structure and species richness will most assumable attracts more/other fauna and flora.

Fauna

See explanation on flora. It is generally assumed that a larger diversity in vertical structure and flora richness will attract more/other fauna species that are currently not present in the area. Some literature mentions a decrease in flora and fauna after establishment of pinus plantations, but these observations has been done in cases where natural forest was replaced by pinus plantations. This is not the case in the project area of Piura. Also research was done in bird species richness in native tree plantations compared to species richness in pinus plantations. Hjarsen, T. (1997) found that endemic bird species were never found in the pinus plantations but only in the plantations with native tree species, such as Polylepis forests. Also insect, fruit and nectar eating species were found in the lowest abundances in the pinus plantations (ref: 04_14_Hjarsen_1997, page 3). On the opposite, seed eating species with large range sizes and broad habitat selection were relatively more abundant in the pinus plantations. On the bird rarity score, pinus plantations score similar as endemic Polylepis forests (ref: 04_14_Hjarsen_1997, page 4, table 1).

According to most literature, fauna richness will almost certainly increase with reforestation, even with reforestation of pinus species. Of course, from an ecological point of view preference is given to reforestation with exclusively native tree species, but in this case a deliberation has been made between socio-economic - and ecological priorities. An increase in the diversity of fauna and the number of animals will probably only be visible many years after plantation establishment. As said, the effects will be limited in the *Pinus* plantations since these exotic tree species do not provide suitable food for most endemic fauna species. In the parts where *Alnus jorullensis* and *Polylepis incana* is planted, fauna richness may increase to a larger extent.

Potential negative aspects of project

Flora

Proper plantation management (e.g. by sufficient thinning and opening the tree canopy) is necessary so that understory vegetation can develop. This is especially important if woody species are already present (see ref: ref:04_04_Hofstede et al_2002). If management is lacking, tree canopy will remain closed and soil litter will accumulate, which prevents flora species to develop.

Fauna

According to Albarino (ref: 04_10_Albarino_R._date_unknown, page 1 §2) monospecific plantations of exotic conifers, are expected to affect input and conditioning of organic matter as well as the quantity and quality of this resource with consequences for the detrital food web. Previous evidence (field and laboratory experiments) has shown that leaf litter of the dominant native trees are processed significantly faster than pine needles and that detritivores (invertebrate shredders) prefer to elucidate native broadleaves to pine needles.

	Mitigating activities of negative aspects **Flora** The plantation areas will be reforested with both pines species as with native broad leaved tree species (*Polylepis incana* and *Alnus jorrulensis*). The pine species are selected because these species are preferred and supported by the local population. During to first rotation, proper management will be carried out in collaboration with the local farmers. By doing so, tree canopy will open up within a 5-7 years after the project start and the accumulation of litter will be prevented. Hence, proper management of the pinus plantations is essential to prevent the competition and suppression of understory flora species by the pinus trees. This is also true for the native tree species, but to a lesser extent. Nevertheless, also the Alnus plantations will receive proper management (thinning, pruning). After the first rotation cycle a conversion management in needed in which the percentage of native tree species in the project area will gradually be increased. **Fauna** See comment on flora. Proper management of the plantations (thinning, pruning) will prevent the accumulation of litter and will stimulate the development of understory flora.
Micro climate	**Current situation** **Temperatures** Average yearly temperatures in the project area are 10.7 ºC (ref: 04_13_Climate-diagram for Choco) **Precipitation** Average yearly precipitation in the project area is 598-750mm (ref: 04_13_Climate-diagram for Choco) **Positive aspects of project** **Temperatures (soil and air)** Microclimatic variables, particularly solar radiation, air temperature at the ground surface (surface temperature), and soil temperature, are highly sensitive to changes in the overstory canopy and exhibit relatively high spatial and temporal variability within a forest (See ref: 04_19_Microclimate_Chen_1999, page 289, §3) According to Chen et al. (1999) forest temperatures are lower in the day and relatively warmer at night. Mean air and soil temperatures usually reach their minima before sunrise and their maxima in the mid- or late afternoon, depending on geographic location, position in the landscape, and overstory structure. Variabilities in air and soil temperatures are also greater during the day than at night, and variation is greatest in the mid-morning and/or the late afternoon, with twin-peak patterns for temperature and moisture (See ref: 04_19_Microclimate_Chen_1999, page 290, §2). Hence, as soon as the forest will develop, it is assumable that the forest-like environment will buffer extremes in air- and soil temperatures in the project area: Variations in local temperature will be smaller than they are now. The soil microclimate is important for the activities of soil biota and the decomposition of organic material: Soil temperatures (and soil moisture) are strongly related to the decomposition of organic material. The association between decomposition and soil temperature is linear, whereas that between decomposition and soil moisture is nonlinear. (See ref: 04_19_Microclimate_Chen_1999, page 288, §3). **Precipitation** Due to the small scale of the plantations, it is not expected that the plantation will

	have an noticeable effect on the quantity of precipitation. However, dense fog often appear in the area and trees will assumable catch more air moisture from dense fog than pasture vegetation. A lot of this moisture will evaporate before it reaches the ground, but a certain percentage will reach the understory vegetation. Assumable, the plantations will alter to a small extent the precipitation in the area. **Potential negative aspects of project** **Temperatures (soil and air)** No direct negative effects are expected of the reforestation activities on the current surface (soil and air) temperatures. **Precipitation** No direct negative effects are expected of the reforestation activities on the current precipitation in the project area. **Mitigating activities of negative aspects** **Temperatures** Proper management of the plantations is always needed. No mitigating activities are needed since no negative impacts on surface temperatures are expected. **Precipitation** As mentioned before, it is not expected that the proposed reforestation activities will negatively impact the soil water content in the area. In the past (before the 1950's) a large part of the project area was covered with forest vegetation (according to interviews, see ref: 01_01_VIDEOS ELEGIBILIDAD), so reforesting the area seems fairly acceptable in particular because of the erosion problems in the lower parts of the Piura basin.

3. Regarding the biodiversity of a project area, all endangered and critically endangered species of the IUCN Red list must be identified and evidence must be given that appropriate activities are put into place to protect them.

To verify the status of flora species, the following link was consulted:
http://www.iucnredlist.org/apps/redlist/search

Of all the species mentioned above (by Garcia, D., 2007. Ref: 04_12_Garcia_D.etall_2007), species found in endangered and critically endangered in the IUCN red list are:

	Name of specie (English, Latin and if possible local language)	**Status** of IUCN Red list	**Activity to protect the specie**
1	*Chuquiraga*	EN	In case it is found in the project area, expert consultation will be arranged to find optimal protection measures. Local farmers will be informed about the importance of the species and a protection program will be developed.
2	*Calceolaria spp.*	NT/ VU/ EN	In case it is found in the project area, expert consultation will be arranged to find optimal protection measures. Local farmers will be informed about the importance of the species and a protection program will be developed.

Also '*Norma Legal Nª9526, Pages 323529 – 323534'* was consulted to see the situation of flora species.

	Name of specie (English, Latin and if	**Status** of IUCN Red list	**Activity to protect the specie**

	possible local language)		
1	*Myrcianthes discolor*	CR	In case it is found in the project area, expert consultation will be arranged to find optimal protection measures. Local farmers will be informed about the importance of the species and a protection program will be developed.
2	*Polylepis incana (Found outside project area)*	CR	The project activities entail the planting of *Polylepis incana* that will be used for conservation purposes only.

To verify the status of fauna species, the following link was consulted:
http://www.iucnredlist.org/apps/redlist/search

The result of the inventory is that no fauna species mentioned above (by Garcia, D., 2007. Ref: 04_12_Garcia_D.etall_2007) were found in endangered and critically endangered in the IUCN red list.

4. Evidence must be given that the use of chemical products (such as herbicides and insecticides) is minimized in its application.
Currently sheep manure is used for preparation of the soil substrate in the nurseries.

Potential products used for project activities:
The products in the table below may be used for the project activities. All products listed in the table below will be used in the nurseries and will be used in minimum quantities. All other potentially used chemical products or fertilizers, not listed in the table below, will also be used in minimum quantities. Thorough considerations will be made by the project management regarding the necessity of their use.

	Name of product	**Main ingredient**	**Purpose of use**
1	Nitrofoska (‘Abono Triple’)	N, P, K in equal quantities	Nursery: Fertilizer for the soil substrate in nursery
2	Homai	Thiophanate methile + Thiram	Pesticide (preventive use) against a fungus ‘Chupadera’ (*Rhizoctonia solani, Fusarium sp. and Pythium sp.*) that attacks tree juveniles and smaller trees in the tree nurseries. This substrate is important to prevent attacks of these fungus because in the past these fungus has almost entirely killed all tree juveniles in the nurseries of caserios Cajas and Huambichi.

5. Evidence must be given, that the application of chemical products is documented.
It is essential that control and documentation on the use of pesticides and other chemical products is adequate and transparent. Illegal and improper use of chemicals can endanger the regular and intermediate certifications of the project. As such, the use of chemical products shall be documented and clear evidence shall be provided to the certifier in case requested. **Documentation of chemical products** The project manager is responsible for the proper use and documentation on the use of chemical products. In all circumstances and at any time, he/she must be able to present clear documentation and evidence in order to provide full transparency to the certifiers. The following data will be documented: • Receipts of the supplier of the chemical products. The receipts must clearly indicate the name of the supplier, the date of delivery, the name of the chemical product and the amount of the product that has been purchased (L or Kg) • Amounts and name of chemicals that have been used in the field (Liters or Kg) • Amount of chemicals (L or Kg) that are still in stock (in the depository). • Number of hectares that have been treated with the respective chemical • Management units ID's that have been treated with the respective chemical • Time period in which the management units have been treated with the respective chemical • Argumentation why the respective management units have been treated with the a chemical The project manager must also be able to provide evidence to the certifiers that the chemicals used are not ecologically harmful for the environment.

6. Evidence must be given, that when chemicals are used, sufficient training and proper equipment is provided to minimize environmental impacts.
In case chemical products will be used for the project, the relevant Material Safety Data Sheet will be used for the respective substance. A Material Safety Data Sheet is a document that provides information about a hazardous substance, how it should be used and how to avoid harm when using them. Manufacturers, importers or suppliers of any hazardous substance must provide sufficient information so that the substance can be used safely. The fieldworkers that need to work with the chemical substance, will receive all the necessary protection (clothes, gloves, face/mouth protection, etc.) and instructions. Photos will be made during the fieldwork, which will support the above mentioned argumentation. Source: http://www.commerce.wa.gov.au/Worksafe/Content/Industries/Agriculture,_forestry_and_fish/Further_information_/Agriculture_workbook/Agricultural_chemical_safety_Pesticides.html

7. Evidence must be given that waste is disposed of in an environmentally appropriate way.

The populated centre of Choco and the other caserios in the project area doesn't have a waste management system. Organic material is being re-used by the farmers. Other non-reusable materials are simply being burned or throwed away in from a hill close to the first aid station.





Photo 3 and 4: Garbage in Caserio Choco

The project may generate various types of non-recycable garbage (e.g. plastic bags for seedlings, packaging materials, etc.) and maybe even chemical liquid residues (pesticides). A clear and environmental friendly strategy is needed for the disposal of this garbage. The waste disposal strategy will also serve for awareness raising amongst the people living in the project area, since many of the project employees will be local farmers. However, the waste disposal strategy mentioned below will focus on the waste that is generated by the project, not by the entire population in and around the project area. Every project employee (inluding the project manager) should respect the waste disposal rules of the project. All waste that is in one way or another related to the project activities need to be processed according to the established rules. The project manager and the project coordinator will have the responsibility to verify if these rules are obeyed and complied with.

The following actions are foreseen to be taken [(1)]:

1. **Awareness raising amongst the project employees**

2. **Separation of garbage.-** Awarenes should be created amongst the project employees regarding the quality and composition of the garbage to enable them to classify the garbage according to organic garbage (decaying) and non-organic garbage (not decaying).

   The project workers need to be learned to classify the garbage in:

   - Iron material (old cooking pans, cans, old cooking material and old dish, etc.)
   - Glass (bottles, broken cups and mirrors, etc.)
   - Plastics (Bottles, plastic cups, plastic packing material, etc.)
   - Paper (news papers, magazines, paper boxes, etc.)
   - Dangerous or very unhygienic garbage (used bandage from wounds, old medicines, toilet paper, used diapers, meat, old batteries, etc.)
   - Organic material (food remains, remains of harvests, old egg shells, etc.)

3. **Preperation of compost.-** The project employees will be learned how to make compost from collected organic material. To do so, the organic material should be shredded to size of maximum 10cm. This material should then be mixed with manure. This mixture must then be stored in a humid and dark place and be covered with material to prevent nuisance from flies and mosquitos. At a later stage the compost can be used for the project again.

4. **Identification of recyclable residues.-** The project employees will be trained in recognizing the differences between recycable redidues and non-recycable residues.

5. **Preparation of locations to store garbage.-** Ideally six (6) storage places should be established (large buckets or plastic containers/sacs), where all the project waste is being collected. Each storage will contain a different class of disposal (see classes above).

6. **Location of central collection point**,- Ideally a central collection point should be established in the centre of Choco to collect all non-recycable material generated by the project activities (batteries, medicines, plastic bottles, etc.)

7. **Coordinate garbage collection with the local authority.-** An agreement should be made with the local authority that the non-recycable material will be transported/removed from central collection point in Choco to the nearest garbage dump where the garbage can be processed in a considered (and preferably environmental friendly) manner.

(1) The above mentioned methodology is based on the experience that ONG PIDECAFE gained with the community La Capilla (12000 masl). This community is located in the district San Miguel del Faige, Province of Huancabamba, Department of Piura.
The project management will offer also some basic training/education to the villagers to create awaremess regarding the management of waste.

8. Evidence must be given, that 15 meter wide buffer strips along permanent or temporary watercourses (streams, rivers, wetlands) are implemented. These buffer strips must be
   - part of the nature conservation area, OR
   - must be managed according to '06 CO_2-fixation - Option 1b) Conservation forest'.

   In both cases only native tree species are allowed to be planted.

In the plantation areas, no major watercourses are present. Only a six (6) small water pools and an irrigation channel have been found. A buffer of 15 meters will be preserved along these water bodies.

9. Evidence must be given that no flooding irrigation, regular irrigation or drainage is introduced on the project area.

No flooding irrigation, regular irrigation or drainage will be used in the plantation areas of the project Reforestation Sierra Piura.

10. In case of forest operations (thinning, harvesting of timber or non-timber products, etc.), evidence must be given that minimal soil disturbance occurs.
No large and heavy machinery will enter the plantation sites because: 1. Such machines are not available in the area 2. The terrain's topography (steep slopes) doesn't allow entrance of large and heavy machinery 3. The scale of exploitation is too small to make the use of large machinery cost-effective Thinning and pruning of the trees will be/is carried out by hand, with manual equipment (Pruning saws, limbing saws, chain saws, etc). The tree branches and logs will be transported by horses/donkeys. The final harvest will also be carried out with manual tools (chainsaws). Transportation of the logs from the site to the truck road will be carried out by horses/donkeys. In case it is necessary, a temporary truck road will be created for the transportation of the timber during thinning and the final harvest.

11. Evidence must be given that no area-wide ploughing is applied and that mechanized ploughing is limited to the purpose of planting.
No area-wide ploughing will be applied, nor mechanized ploughing. The small trees will be planted individual planting holes (40x40x40cm). In case necessary, the field workers will be trained in appropriate planting techniques (use appropriate planting distance, removal of vegetation, proper positioning of tree roots in the soil). Photos of the planting activities will be taken to support evidence for the regular certifications.

12. Evidence must be given that no genetically modified species are being used.
The source of the planting material will be documented. In case the seeds are purchased from a commercial seed supplier, the following data will be collected: • Contact details of the seed supplier • Declaration of supplier source of origin of seeds In case the seeds are collected by the project workers the following data will be collected: • Photo and/or film during the collection of the seeds • GPS points of the location of the source • Declaration and of responsible person The first 24 ha of the project will be planted (in 2010) with planting material that is already present in the nurseries. The seed that was used for the present nurseries origins from the municipality of Yamango. This municipality purchased the seed from commercial resellers in Lima. This seed was certified. The remaining seeds for the plantation will be purchased at commercial seeds banks in Peru. In Peru no commercial seed banks are present that sell genetically modified seeds. The project management and the project developers hereby declare that no genetically modified seeds will be used or have been used in the plantations.

13. Evidence must be given that the project management is planting native species in mixed stands and in case the timber of the forest is being used, selective harvesting management is applied.

Otherwise, the project must justify its
- choice of tree species
- silvicultural system
- harvesting method

The plantation areas will be reforested with four tree species:

1. **Pinus patula (exotic)**
   This tree species is very tolerant to low temperatures, strong winds, acidic soils and high altitudes.
2. **Pinus radiate (exotic)**
   This tree species is very tolerant to low temperatures, strong winds, acidic soils and high altitudes.
3. **Alnus jorullensis (native)**
   This tree species is tolerant to low temperatures, acidic soils and high altitudes. This species should not be planted on mountain ridges or other areas with strong winds. Best places are along river sides, shallow sites, sites protected from wind and erosion gullies.
4. **Polylepis incana (native)**
   This species is tolerant to low precipitation. It is tolerant to slightly acidic soils (pH between 3.9 and 7.2). Not to be planted on windy locations.

Justification of tree species
For the first rotation, the species *Pinus patula* and *Pinus radiata* has been selected. These species have the preference of the local farmers because it has already been successfully planted in other parts of the Andes at similar harsh growing conditions (high altitudes, poor/moderate soils, strong winds). The farmers are familiar with the tree species and fairly confident with its management. As such, the use of *Pinus patula* and *Pinus radiata* has the full support of the farmers, which is essential for the social acceptance and success of the project.

There are no other pioneer (native) tree species that assure similar growing rates at similar growing conditions. Nevertheless in close consultation with the farmers, the project management has decided to use two native tree species: *Alnus jorullensis* and *Polylepis incana*. Namely, it is assumed that a wider diversity in tree species will contribute to a higher biodiversity: Some mammals and birds will probable be more attracted to native tree species than to exotic species. However, the use of the proposed natives tree species will have its impact on wood volumes. *Polylepis incana* is slow grower and although its high wood density, this species will fixate less CO2 compared to the other tree species. *Alnus jorullensis* is a fairly fast grower compared to *Polylepis incana* but technical experience with the management of this tree species is not very extended yet in the area. Hence both native tree species introduce a certain economic risk, not only for the project investors, but also for the community involved. By using a proper mixture of exotic and native tree species, this economic risk will be reduced while on the other hand contributing to ecological requirement to a certain extent.

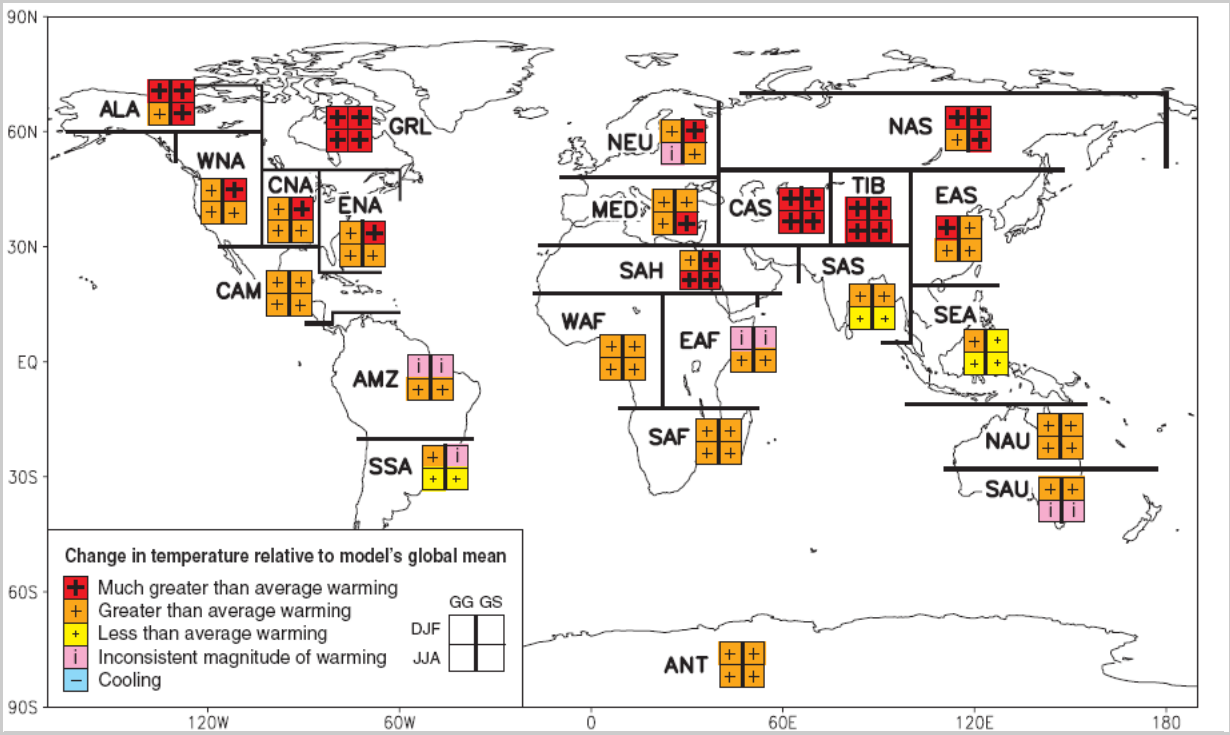
After the first rotation (25 years) the objective is to carry out a conversion management strategy in which the percentage of exotic tree species will be gradually replaced by native tree species. The appropriate strategy and suitable native tree species for this conversion will be studied and negotiated with the farmers during the first rotation period.

Management
The areas will be subdivided into several forest management units (see ref: 03_09_Plantation_management_units.pdf). Each management unit distinct itself from other units regarding the growing conditions: soil type and/or slope degree and/or slope direction, expected degree of wind, presence of soil moisture. Also the distance to erosion prone locations will be taken into account. Based on the growing conditions for each tree species and the site preferences of the farmers, a management unit map was established. Each management unit will be planted with one unique tree species (see tree species above). As a result, the entire plantation area will be a mixture of (four) tree species.

During the rotation cycle, several thinnings will be carried out (1 to 4, depending on the tree species) for the commercial tree species (*P. patula, P. radiate and Alnus jorullensis*). Also prunings will be

carried out (1 to 2, depending on the tree species). At the end of the rotation cycle, a majority of the trees will be gradually harvested in a few year time (except the *Polylepis incana* stands). However, no clear cut will be carried out: A certain percentage of trees will remain standing to stimulate natural regeneration. The project management will aim to a type of forest conversion management that gradually convert the forest plantation to a forest with a higher diversity in tree species and fauna. This may be achieved by actively replant the areas with new (native) tree species or by natural regeneration of abundant tree and plant species. The most appropriate forest management to carry out/stimulate such gradual conversion will be developed during the runtime of the project.

14. Evidence must be given that all species planted are site-adapted under changing climate conditions.



Source: www.grida.no/climate/ipcc%5Ftar – Report 'The Scientific Basis' – Chapter 10 – Page 596

In 2005, a climate change scenario study to 2050 was carried out for the Piura basin, of which the project (Reforestation Sierra Piura) forms part (see ref: 04_15_CONAM_climate_change_and_adaptations). The study was commissioned by the Consejo Nacional del Ambiente (CONAM) and was carried out by SENAMHI (The National Service of Climate and Hydrology in Peru). Six climate change scenarios were investigated that each depend on the decisions that mankind takes at present (ref: 04_16_CONAM_climate_change_scenarios).

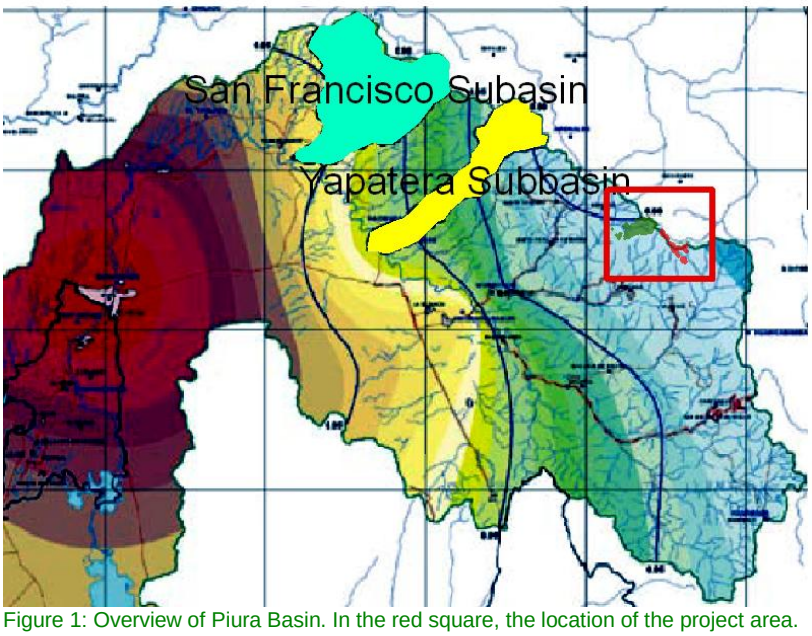


Figure 1: Overview of Piura Basin. In the red square, the location of the project area.

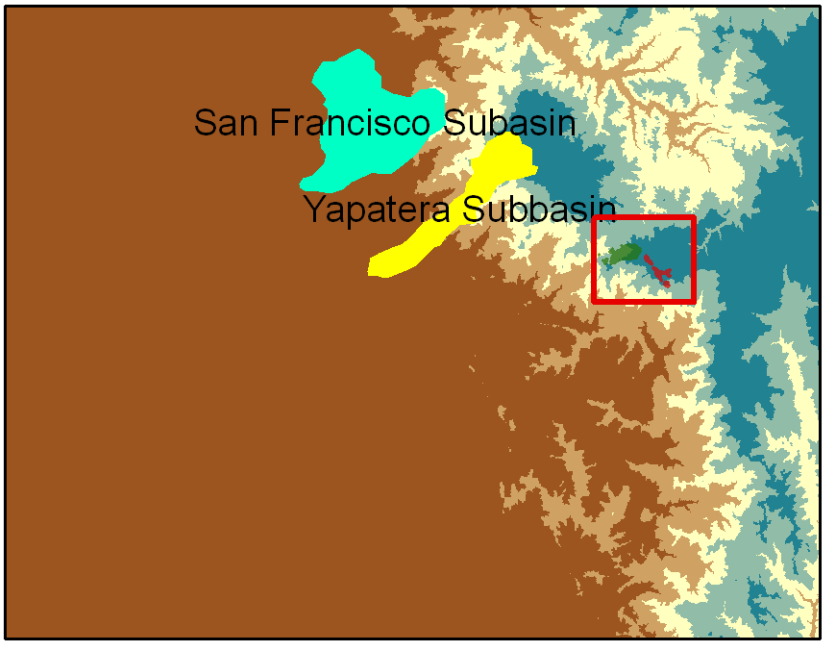


Figure 2: San Francisco Basin and the Yapatera Basin plotted on an elevation background. In the red square the location of the project area.

As can be seen in figure 2, elevations in the higher part of the Yapatera Subbasin are comparable with those of the project area. As such, it can be expected that the expected climate change in the higher part of the Yapatera Subbasin will be similar with that of the project area.
According to CONAM/SENAMHI climate study (ref: 04_16_CONAM_climate_change_scenarios, page 18, §3), the expected rainfalls for the higher parts of the Yapatera river sub basin, would exceed in a **390mm** average, whilst in the lower parts there could be a deficit in rainfalls up to 25 mm, for which the

water balance would exceed its normal values in the higher parts of this sub basin, and there could occur a surplus of up to 350 mm.

According to the historical climate data from the climate stations Arenales (3200m asl) and Chalaco (2200 m asl) (see ref: 04_17_Map_locations_of_whether_stations), **actual average precipitation** in the project area is estimated to be around **598-750mm/year** (average value between weather station at Arenales and at Chalaco). In case of a worst case climate scenario, precipitation will have a surplus of 350-390mm in the higher parts of the Yapatera subbasin, which mean that by the year 2050, the average annual precipitation would be around **1100mm/year.**

According to the historical climate data from the climate stations Arenales (3200m asl) and Chalaco (2200 m asl) (see ref: 04_17_Map_locations_of_whether_stations), the average temperature in the project area is estimated to be around 12 – 14 ºC (average value between weather station at Arenales and at Chalaco). The expected temperature rise will be a few degrees. So by the year 2050, average yearly temperature may rise to 15-18ºC.

Below, the maximum temperature and precipitation for each tree species is indicated. As can be seen, all the trees will withstand the anticipated increases in rainfall and temperature. So taking these empirical prognosis into account, the plantations will survive by the year 2050.

Pinus patula
Max temperature: 29ºC
Max precipitation: 2000mm

Pinus radiate
Max temperature: 30ºC
Max precipitation: 1600mm

Alnus jorullensis
Max temperature: 27ºC
Max precipitation: 3000mm

Polylepis incana
Max temperature: unknown, but naturally occurs at temperatures around 8-9 ºC.
Max precipitation: >500mm (aridity is more limiting than precipitation at these elevations)

05 Socioeconomic Aspects

Name of the Project	Reforestation Sierra Piura

1. Evidence must be given that the project has net-positive socio-economic impacts.

   Therefore,
   a. positive socio-economic impacts of the projects must be enhanced and
   b. negative or potentially negative socio-economic impacts must be mitigated.

   For point a. descriptions of the following aspects must be given:
   - Creation of employment
   - Capacity building
   - Welfare activities

   For point b. descriptions of the following aspects must be given:
   - Stakeholder involvement prior to the project start
   - Displacement of people
   - Spiritual, religious, or other socially important places influenced by the project activities

Creation of employment

	Amount of people (currently: 1st year) Planting of 24ha and improvement of nurseries		**Amount of people** (year 2) Planting of 94ha/year, management of nurseries		**Amount of people** (year 3 →) Planting of 106/year, management of nurseries	
Management	1 part-time (2 days/week)	0 seasonal	1 part-time (2 day/week)	0 seasonal	1 part-time (2 day/week)	0 seasonal
Employees (project coordinator)	1 full-time (5 days/week for project coordination)	0 seasonal	1 full-time (5 days/week for project coordination)	0 seasonal	1 full-time (5 days/week for project coordination)	0 seasonal
Tecnical Support		2 part time (2.5 days/week)		2 part time (1.5 days week)		2 part time (1.5 days week)
Financial Support (Independant auditor)		1 part time (1 day/month)		1 part time (1 day/month)		1 part time (1 days month)
Field assistants (field support during reforestation, nursery maintenance, and training)	2 full time (5 days/week)	1 seasonal (5 days/week for 6 months)	2 full time (5 days/week)	1 seasonal (days week for 6 months)	2 full time (5 days week)	1 seasonal (5 days/week for 6 months)
Workers (for work in	9 full time (1	45 seasonal (1day)	9 full-time (1	27 seasonal	9 full-time (1	27 seasonal

	nurseries and other supportive forestry activities)	day/week) for forest maintenance	for seedbeds 23 seasonal (3 days/week/2months) for various activities*	days/week) for forestry maintenance activities	(1day) for seedbeds 25 seasonal (3 days/ week/ 2 months) for various activities*	day week) for forest maintenance	(1day) for seedbeds 10 seasonal (2 days/week/ 1 month) for various activities*
	Workers (planting and other forestry activities)	6 full time (2 days /week) for mantainance activities (4p en Choco, 1p en Santa Cruz,1p en las Huacas)	124 seasonal (6 days/week / 3 months) for planting	21 full time (2 days/week) for forestry maintenance activities	401 seasonal (6days/week/3 months) for planting	35 full time (2 days/week) for maintenance work	400 seasonal (6days/week/ 3months) for planting
	SUM	19 full time	196 seasonal	34 full time	457 seasonal	48 full time	441 seasonal

*Various activities: Preparing soil substrates + Filling plastic bags + Transplant small juvenills from seedbeds to plastic bags.

Time estimations:

- Pruning Pinus spp.: 1.5min/tree * 1110 trees/ha = 1665min/ha = 27.75hours/ha = 3.5 man days/ha
- Pruning Alnus: 1.5min/tree * 1480 trees/ha = 1665min/ha = 37hours/ha = 4.6 man days/ha
- Thinning (cutting and pruning and tipping with chainsaws): ± 100 trees/day/person (taking into account steep slopes, altitude and maintenance of equipment) = 11.1 to 14.8 man days/ha.

Capacity building	The farmers will benefit from the project by participating in the project activities. They will be fully informed and fully involved in the project activities. Before the start of the project activities (field work)* a list of activities will be presented to the farmers, including a time frame in which these activities need to be carried out. The activities will be coordinated and planned by the farmers themselves, since they are the people that need to carry out the project. However, they will be supported by technical experts (field managers). If necessary, (basic) training will be given to carry out the activities in a proper way. In January 2010 a chronogram will be made of the planned project activities, that will be discussed with the farmers (see attachments). After this, regular meetings will be arranged between the project management and the farmers, wherein the progress and the follow-up-activities will be discussed. **Until now, only activities have been carried out that are related to the elaboration of this documentation. Also technical assistance has been given to improve the tree nurseries that are present in the project area.*

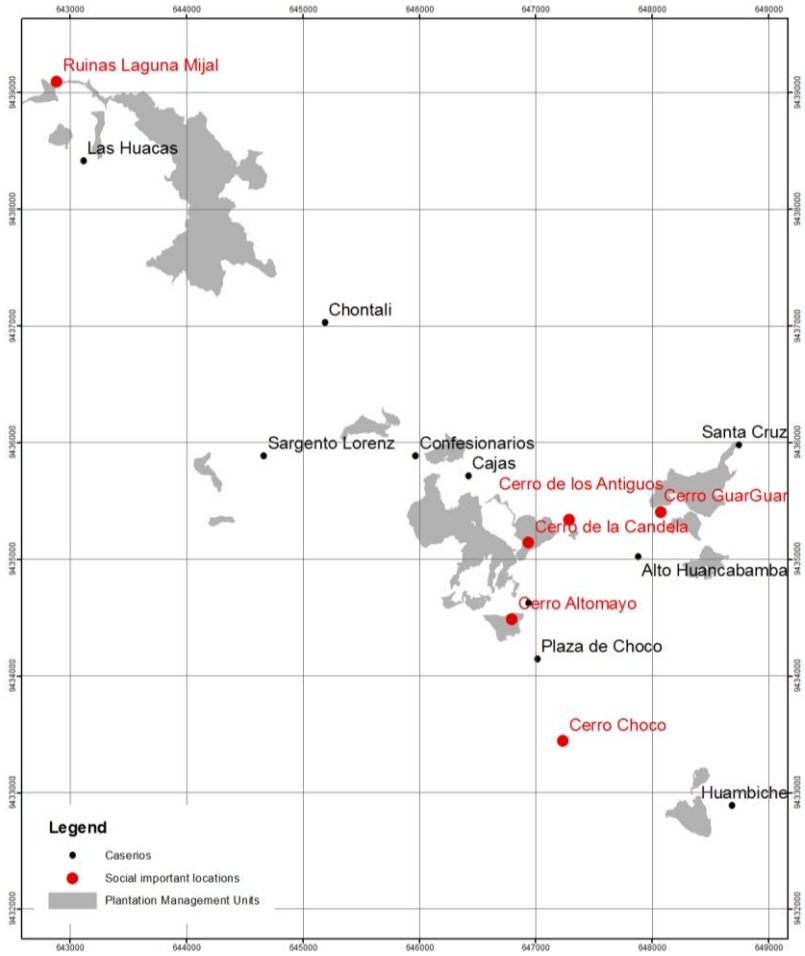
Welfare activities	The project coordinators will organize and coordinate health campaigns in the populated centre of Choco. This will be done in collaboration with CLASS (Centro Local de Atención en Salud), which is a local centre for health care and awareness in Yamango (see Ref: 05_09_Welfare_activities_cooperation)
Involvement prior project start	In June 2008, employees of PIDECAFE (Programa Integral para el Desarrollo del Café), Peruvian ONG that develops projects in the North-East part of the Andes (North Oriental) in Peru) together with VSF-CICDA (Veterinarios Sin Fronteras - Centro Internacional de Cooperación para el desarrollo Agrícola, French organisation that supports farmers in the poor regions) arranged meetings in various districts of the Sierra Piura to select potential areas for a CO2 reforestation project. The inventory concluded that the centre of Choco was the best place to carry out such a project, not only because the project would have full support of the mayor (Mr. Isauro García Nuñez) but also because of the past experiences that the community had with reforestation (ONG Villa Nazareth) and because of the fact that sufficient land was available for reforestation.  Photo 1: Visit at the Caserio Huarequeque, District of Canchaque, Province of Huancabamba, June  Photo 2: Visit at Caserío Huacas, District of Yamango, Province of Morropón, June 2008. In August 2008, field work was carried out in Choco ('Centro Poblado de Choco') to elaborate a Proyecto Idea Nota (PIN) (see reference: 05-12_PIN Sierra Piura.pdf). First a meeting was arranged with the representatives of the 11 'Caserios' in Choco (small communities containing a number of families). After this, arrangements were made for individual meetings with each of the 11 Caserios. It was decided to work with 10 of the 11 Caserios. With 6 Caserios, a meeting was arranged. With the remaining 4 Caserios, informal meetings were held with the some of the representatives. During the meetings, the farmers of the caserios calculated the areas that they were willing to reserve for reforestation activities.

This document will be made publicly available. CO2buyers will be able to use this information for marketing purposes.

	In November 2008, another field visit was carried out. During these meetings, the farmers of the Caserios were informed about the planned reforestation activities. The objective of these meetings was to create a better awareness and understanding about the consequences of the activities among the farmers. For the employees of PIDECAFE and VSF-CICDA these meetings were important to verify if the farmers were fully supporting the planned reforestation activities and were fully aware of the impact that these activities would have on their lives and areas. After having verified this, 'contracts of agreement' were signed by the representatives of each Caserio. During the field work also the existing (old and poorly managed) nurseries were visited and potential plantation areas were visited and mapped by GPS. Also other information was collected to get a better understanding of the situation in the area. In July 2009 all the limits of the plantation areas were accurately mapped by GPS (5-10m horizontal accuracy). This was done in the presence of the representatives of the involved caserios. In August 2009, four (4) extra areas were included and measured by GPS. Also, vegetation samples were taken for the baseline study and for flora determination (plant species). Again, all this was done in the presence of the representatives of the Caserios. In October and November 2009 a number of field visits were carried out to collect more field data. Also interview and meetings with the farmers were made to generate commitment among the farmers and mutual trust between the farmers and the project management.
Displacement of people	The project will be carried out on communal terrain that is legally owned by the project stakeholders (caserios). These caserios have the official land titles according to Peruvian legislation (see ref: 05_11_SUNARP). The foreseeable plantation areas are appointed by the caserio's representatives themselves. No houses or farmland is located on these terrains. As such, displacement of people is not an issue.
Socially important places	The project stakeholders (farmers) in the project area have a number of places that are of importance for them. These are places of archaeological importance and of social importance. Places that are of archaeological importance are: 1) 'Cerro Misal'. Here an archaeological structure of 28x78m is found, subdivided in various symmetric rooms with earthen walls. 2) 'Cerro al Tomayo'. Close to 200 rectangular shaped archaeological constructions are found here. The structures are in very bad condition but clearly visible. Platforms and other remainings have also been found here. 3) 'Cerro de los Antiguos'. Here an archaeological construction is found, like a kind of campus. The site consists of a wide network of narrow channels that are interconnected with each other.

4) ‘Cerro de la candela’. Here numerous platforms and courts are found.

Small archaeological sites have been found in the hills of Choco, Alto Huancabamba, Santa Cruz, Chontalí and Cajas.



Map 1: Social important locations in the project area.

In the planned plantation areas of Confesionarios and Cajas, various small archaeological structures have been found.

	These sites will be respected by all project stakeholders and the project management and have been extracted from the plantation management units. First the outer limits of the archaeological sites have been determined by GPS. An overview of the point locations of these archaeological structures is given in the picture above (see ref: 05_10_Social_Important_Locations). After that a 15m buffer will be created by means of GIS. These buffers will be loaded into the GPS again to determine the buffer limits in the field. The buffer limits will be marked in the field by wooden poles. A shapefile has been generated with the exact coordinates of the archaeological sites in the map 1 above.

2. To give further evidence for the criteria 1a. and 1b. a stakeholder consultation must take place a maximum of 3 months before or during the field visit of every certification process.

   For this consultation, all stakeholders must be invited
   - using appropriate means +
   - providing them with sufficient information on the project activities.

   The results of the consultations must be documented.

	Stakeholder	**Means of invitation** Radio (R), Newspaper (N), Letter (L), Posted notice (Pn), Email (E), Fax (F)	**Information supplied to them** CFS, Project description (Pd), ...	**Feedback from stakeholders**
1	All members of the caserios involved (owner of the land, owner of timber)	R, Mouth-to_Mouth	Description of project plans (forest management activities + chronology) Inform people about the certification procedures and overall project objectives.	Summary of comments and opinions of caserios. List of actions to be made by project management. Again, a confirmation of approval of project activities is asked from stakeholders. Attendance lists of the meetings will be elaborated that will be signed by the participating farmers. For the results of the meetings, farmer feedback and the presentation material that has been provided to the farmers see ref: 05_13_Consultation_and_feedback_farmers.zip
2	CAFEDIRECT (purchaser of first batch of carbon credits)	Telephone + L + E	Description of project + Description of project area and social situation + description of involved parties + description of Carbon Fix Standard	List with comments and actions to be made by all involved parties. Feedback is requested from CAFEDIRECT by email. Feedback of CAFEDIRECT is provided in ref: 05_14_Consultation_and_feedback_particip.zip
3	CEPICAFE (responsible for marketing the carbon credits & co-financer of the project).	T + E	Description of project activities to be carried out + description of Carbon	List with comments and actions to be made by all involved parties. Feedback by email.

			Fix Standard. People from NORANDINO should be well informed about their tasks and responsibilities	Feedback of CEPICAFE is provided in ref: 05_14_Consultation_and_feedback_particip.zip
4	COOP NORANDINO / PROGRESO (technical management of project and project owner)	T + E	Description of project activities to be carried out + description of Carbon Fix Standard. People from NORANDINO must be well informed about their tasks and responsibilities. Emphasize must given on the importance of obeying the Carbonfix criteria. Project management must be well informed about the regulations that have to be followed and obeyed.	List with comments and actions to be made by all involved parties. Feedback by email. Feedback of PROGRESO is provided in ref: 05_14_Consultation_and_feedback_particip_organisations.zip
5	Forest authority of Peru (Administracion Tecnica de Fauna Silvestre Tumbes – Piura)	T + E + L	Information will be provided about the project plans, the elaboration process, the support of the communities and the Carbon Fix Standard	Feedback from the Forest Authority expected. A summary of this feedback will be written and send to all attendants by email. Feedback of ATFFS-TP is provided in ref: 05_14_Consultation_and_feedback_particip_organisations.zip A request has been submitted by ATFFS-TP that the plantation should be registered as soon at it has been established. This will be taken care of.
6	National NGOs (e.g. University of Piura)	T + E + L	Information will be provided about the project plans, the elaboration process, the support of the communities and the Carbon Fix Standard	Feedback from the NGO's expected. Feedback of the University of Piura (UNEP) is provided in ref: 05_14_Consultation_and_feedback_particip.zip

3. Evidence must be given that first aid kits are reasonably accessible for the working staff.
At the management office a first aid kit will be present with the most elementary articles for physical well-being. The project manager is responsible for the presence, access and maintenance of this kit. This kit will be more sophisticated than the field workers' first aid kid. During field work (planting activities, pruning and thinning, etc, etc.) the project coordinator and field manager (contractor) will be responsible for the presence of a basic first aid kit (transportable). This person will also be trained in providing first aid to the field workers (reanimation, bandage of wounds, artery bleedings, etc.). The field manager is responsible for the presence, access and maintenance of this first aid kit.

4. Evidence must be given that working staff is able to organize itself and voluntarily negotiate with their employers.

The project working staff are the stakeholders (farmers) that are living in the project area. These farmers all belong to one of the ten caserios that are participating in this project (see table 1 below).

Every single caserio has a Reforestation Committee. Hence in total there are ten (10) Reforestation Committees. Every Reforestation Committee is composed of: 1 President, 1 Secretary, 1 Treasures and 2 Spokepersons. These positions are occupied by people from the respective caserio.

Every Reforestation Committee (10) arranges meetings with its constituency (all members of the respective caserio). In these meetings problems, preocupations and dissatisfactions regarding forestry issues are discussed. In case complaints or dissatisfaction would arise among the members of the caserios regarding the project activities, these isues will be discussed in the committee meetings. After internal discussion, the President of the respective Reforestation Committee will report the dissatisfaction to the project manager, who will take the appropiate measures (solve the issue of arrange a meeting with all Reforestation Committees).

The following table provides an overview of the representatives of these committees:

Caserío	**Name of the representative of the Reforestation Committee**
Huambiche	Solano Guevara García
Choco	José Santos Huamán García
Alto Mayo	Fidel García Bermeo
Alto Huancabamba	Carlos Saldaña García
Santa Cruz	Pedro Javier Jiménez García
Cajas	Alfredo Huamán Peña
Confesionarios	José Gabriel Mondragón Córdova
Chontalí	Evaristo Peña Mondragón
Las Huacas	Renelmo Huamán García
Sargento Lorenz	

Table 1: Caserios and their representative in the forestry committee.

5. Evidence must be given that all equipment (tools, machines, etc.) of the working staff are in safe working mode.

All working equipment (including manual equipment) will be regularly (at least each month during forestry activities) checked by a responsible person (field manager). Basic checklists will be established for all hand tools that will be used in the project activities. For the chainsaws a more elaborated checklist will be established. People that has to work with the chainsaws (during thinning and final harvest) will receive appropriate training in safety measures. A checklist (provided by the supplier of the chainsaws) for appropriate maintenance of the chainsaws will be elaborated. The project coordinator is responsible for observance of the established safety rules. Any accident should be clarified and reported by the field assistant to the project coordinator. The project coordinator will inform the project manager about the issue and look for appropriate measures to prevent similar accidents in the future. Safety measures will be developed as much as possible –and if applicable- in accordance with the manual 'Safety and Health in Forestry Work', published by the International Labour Organization (ILO) (see Ref: 05_02_Safety_and_Health_in_Forestry_work).

Checklist protective clothing

During felling, pruning (only above 2.5m) and thinning activities, safety helmets, gloves and goggles will be supplied to the workers that work with hand tools. The chainsaw operators will work with special protective clothing (chainsaw clothing) that is prescribed by the chainsaw supplier as well as the International Labour Organization (ILO). These materials must be in good condition. Safety helmets, gloves and goggles will be checked each month. The chainsaw clothing will be checked each week by the field manager. Clothing that has been torn will be repaired immediately.

Checklist hand tools

The following hand tools will most often be used in the fieldwork: Shovels, hand saws, machetes, pickaxes. In case tools need to be purchased, the following criteria will be taken into consideration (see Ref: 05_02_Safety_and_Health_in_Forestry_work, page 33)

Hand tools purchase

- Hand tools for cutting and splitting should be manufactured from good quality steel which maintains its cutting edge and effectiveness with the minimum amount of maintenance.
- The head of a tool for cutting and splitting should be fixed securely onto the handle with an effective device, for example a wedge, rivet or bolt.
- Handles should provide a secure grip and should be made of good quality wood or other materials suitable for this purpose.
- The specification of tools, such as size, length of handles and weight, should be appropriate to cater for the needs of the work and the physical attributes of the user.
- When not in use, sharp-edged tools should be sheathed with an appropriate device.
- All tools that are purchased from the project budget will be labelled or marked/tagged with an unique ID. Tools that are used in the project activities, but are not in ownership of the project will be named at the owner of the tool.

Hand tool checkslist (at least each month)

The condition of the hand tools will be checked each month by the field manager according to the table below. In case a failure is found, the tool may not be used in the activities before it has been repaired.

ID of tool / Name owner of tool.	Tool	Item to be checked	Check (good/bad)	Date of check
	Shovel	Shaft strong		10 Oct 2010
		Sharpness of blade		
	Handsaw	Sharpness		
		Metal fatigue		
	Machete	Sharpness of blade		
		Handle fixed		
		Metal fatigue		
	Pickax	Shaft strong		
		Handle fixed		
		Sharpness		

Basic checklist for hand tools.

Checklist chainsaws

In case chainsaws will be utilised for thinning and final harvest, the maintenance and safety rules advised by the supplier will be respected. Every renowned chainsaw supplier (STIHL, Husqvarna, etc.) delivers a manual for maintenance of the equipment. Attachment *Ref: 05_01_Manual_maintenance_chainsaw_STIHL* is an example of such a manual. These manuals are also available in the Spanish language. The workers will be trained in the use of the chainsaws (if people do not have the expertise yet). This training also includes a module for maintenance of the chainsaws. Below a maintenance table is plotted that is extracted from the STIHL manual (*Ref: 05_01_Manual_maintenance_chainsaw_STIHL, page 25*). This maintenance table will be used each day before the work starts and after finishing the work.

Please note: Different models may have different parts and controls.		before starting work	after finishing work or daily	after each refueling stop	weekly	monthly	if faulty	if damaged	as required	see page:
Complete machine	Visual inspection (condition, leaks)	x		x						
	Clean		x							
Throttle trigger , throttle trigger interlock, Master Control, depending on model	Check operation	x		x						
Chain brake	Check operation	x		x						
	Clean by STIHL dealer								x	
Filter in fuel tank	Check					x				
	Clean, Replace filter element							x		
Fuel tank	Clean					x				
Chain oil tank	Clean					x				
Chain lubrication	Check	x								
Saw chain	Inspect, also check sharpness	x		x						
	Check chain tension	x		x						
	Sharpen								x	
Guide bar	Check (wear, damage)	x								
	Clean and turn over				x		x			
	Deburr				x					
	Replace							x	x	
Chain sprocket	Check				x					
Air filter	Clean	x					x			
	Replace							x		
Cooling inlets	Clean		x							
Cylinder fins	Clean					x				
Carburetor	Check idle adjustment – chain must not turn	x		x						
	Readjust idle								x	
Spark plug	Replace						x			
All accessible screws and nuts (not adjusting screws)	Retighten								x	
Rubber vibration buffers	Inspect				x					
	Have replaced by STIHL dealer							x		
Muffler	Inspect					x		x		
	Inspect spark arrestor screen	x								
	Clean or replace spark arrestor screen							x	x	
Chain catcher	Check	x								
	Replace							x		

This document will be made publicly available. CO_2buyers will be able to use this information for marketing purposes.

Important note: When chainsaws are used, the design of chain-saws should comply with international regulations on safety features, such as the European Committee for Standardization publication "European Standard EN 608: Agricultural and forestry machinery – Portable chain-saws – Safety" (Brussels, 1994). Most common chainsaw suppliers do comply with these rules (STIHL, Husqvarna, etc.)

6. Evidence must be given that proper protective equipment and training of the working staff is enforced - especially when chemicals are used.

All appropriate safety measures and protective equipment will be used (according to what is available in Peru) to safeguard the health of the field workers. For most dangerous products used in the project (chainsaws, chemical substances) the product supplier most often prescribes certain safety rules by means of user manuals. These instructions will be respected and established within the working procedures by means of checklists. Also (basic) training and instructions will be given to the people that have to carry out the dangerous work. Photos will be taken during the field work (including the date of the pictures) to support prove of conformity with the safety rules.

Safety measures when using chemicals

For safety measures using chemicals, the guidelines in the ILO manual '*Safety and health in the use of agrochemicals: A guide*' will be used as a reference (Ref: 05_04_ILO_manual_Save_Use_on_Agrochemicals), but only as far as they are applicable in the project area. When using with chemicals, minimum requirements will be complied with such as wearing protective cloths, head gear/respiration protection, goggles and gloves. These minimum requirements will be enforced by the project manager, who is the final responsible in case injuries occur due to a lack of proper safety equipment. Safety rules imposed by the supplier of the chemical product will always be respected and will overrule the minimum requirements or the ILO manual.

The project manager will have a hardcopy of the ILO manual (Ref: 05_04_ILO_manual_Save_Use_on_Agrochemicals) at his/her disposal (in the office) in case of emergency or injuries.

During the use of the chemicals, photos will be taken to prove conformity with the procedures described above.

Safety measures for operating the chainsaw

In case chainsaws will be used, the maintenance and safety rules advised by the product supplier will be respected. Every renowned chainsaw supplier (STIHL, Husqvarna, etc.) delivers a manual that explains how to work safely with the equipment. Attachment *Ref: 05_01_Manual_maintenance_chainsaw_STIHL* is an example of such a manual. These manuals are also available in the Spanish language. It is foreseen that STIHL chainsaws will be purchased (preference of project management). The chainsaw workers will be trained in the safe use and proper maintenance of the chainsaws. The project management will enforce the use of chainsaw clothing during work with a chainsaw: It will be prohibited to work with a chainsaw without wearing protective chainsaw clothing.

During the thinning and harvest activities, photos will be taken to prove conformity with the procedures described above.

Safety measures during transport of timber logs by draught animals
For safety measures and working guidelines, the following ILO guidelines will be respected as far as they are applicable in the project area (see Ref: 05_02_Safety_and_Health_in_Forestry_work, page 78). The following minimum rules will be complied with:

- Only animals with sufficient strength and endurance to cope with the strain of extraction work should be used.
- The animals should be fed, watered and rested according to their physical needs. Only persons who are familiar with the animals' needs and behaviour should work with them.
- Undergrowth on animal trails should be cut as close to the ground as practicable, and obstacles should be carefully removed.
- Extraction should be synchronized with cutting as much as possible, and commence at the furthest point of the extraction route to avoid having to travel over branches, tops and other debris.
- Preferably a suitable harnesses should be used to avoid injury and reduce physical strain on the animals when they are pulling the load.

During the transport of timber logs, photos will be taken to prove conformity with the procedures described above.

Safety measures for road transport of logs
For safety measures and working guidelines, the following ILO guidelines will be respected as much as possible (see Ref: 05_02_Safety_and_Health_in_Forestry_work, page 88) as far as they are applicable. The following minimum rules will be complied with:

- The gradient of roads should, wherever practicable, not exceed 10 percent.
- All truck transport should be carried out in a manner that will minimize road damage. Transport operations should therefore be tailored to the bearing capacity of the road.
- Drivers should: (a) hold the legally required licence appropriate for the type of vehicle they are operating; (b) observe Peruvian national traffic regulations at all times; (c) have a thorough knowledge of the regulations and instructions for operating the particular type of vehicle they are driving; (d) be able to carry out routine servicing and minor repairs; (e) have the final responsibility to ensure that the truck is loaded correctly and securely and that the truck is not overloaded (common sense and operator experience should be used).
- Timber trucks should be sturdy and reliable.
- The trucks should be equipped and in a roadworthy condition, to comply with the requirements of national laws or regulations concerning road safety.
- Workers should keep well clear of swinging or falling logs and outside the risk zone in the event of a fractured cable.
- Vehicles should not be overloaded. There must be strict adherence to traffic laws concerning maximum loads.
- Riding on any part of a timber truck other than in the cab should be strictly prohibited.

During the transport of timber logs, photos will be taken to prove conformity with the procedures described above.

This document will be made publicly available. CO_2buyers will be able to use this information for marketing purposes.

7. Evidence must be given that no children under the age of 16 are working for the project.
The project management will prevent that children under age of 16 will work for the project. This can be guaranteed by the 'Documento Nacional de Identidad (DNI)', which is a compulsory and legal identification for Peruvians older than 18 years. However, also people younger than 18 can request for a DNI. To guarantee that no persons younger than 16 will work for the project, the project management will demand a DNI identification for every person (full-time, part-time and day workers) that wants to work for the project. For full-time and part-time (seasonal) employees working contracts will be arranged. A copy of their DNI will be included in the contract (by photocopy or by taking a photo with a digital photo camera). The contract will be signed by employee. For day workers, only a basic day/week-contracts will be arranged that specifies their salary, full name(s) and age. The contract will be signed by worker. The workers shall indentify themselves with a DNI. The DNI will be documented by taking a photo with a digital photo camera. All working contracts, DNI's and photos will be administered by the project manager or project coordinator.

8. Evidence must be given that contracts are clearly define and include the following aspects:	
For employees	**For contractors**
a. working hours and leave of absence	a. tasks (quantity, quality, time)
b. duties	b. payment
c. salary	c. modalities on the termination of the contract
d. modalities on health insurance	
e. modalities on the termination of the contract	
Contracts for full-time and part-time employees will include at least the following items: a, b, c and e. Modalities on health insurance will depend on the arrangements that can be made with health insurance companies in Peru. The price/quality of such a health insurance and the financial capacity of the project will be the determining factors in such an arrangement. *Ref: 05_07_Contract_Project_coordinator*, shows the contract that has been arranged with the project coordinator. Day-workers will only receive a basic day-contract with the following details: Full name(s) of the employee, day salary received. The same safety rules of the project that apply to full and part-time employees will also apply to day-workers. Ref: *05_06_Payments_for_Labour* shows the evidence that such contracts have been used for the plantation activities. It is not foreseen to use contractors in the project activities. Contractors may only be used for the final harvest. In case contractors will be used, their contracts will include the above mentioned items: a, b, c.	

9. Evidence must be given that working staff is preferably employed from areas close to the project.
The project employees (field workers) that will work for the project will be recruited in the project area (centre of Choco) (see Ref: 05_06_Payments_for_Labour). Further evidence can be obtained by interviews of the certifier with the field workers. It is important to mention that during the development of this project and the negotiations with the local famers (Caserios) a '2x1 salary' agreement was made that the local farmers (field workers) of Choco. This agreement implies that the local farmers of Choco will work 2 days against a salary and 1 day without being paid (called 'famer support').

10. Evidence must be given that throughout the project activity project stakeholders are able to address their concerns to the management staff.
Concerns about any type of project activity can be directly expressed to the project management by any stakeholder in the project area. It is expected that this will be done by the Presidents of the Reforestation Committees and by the representatives of the Caserios. Evidence of such a concern and/or participatory processes in which concern can be expressed to the management staff is provided by Ref: 05_08_Evidence_ participatory_agreements. The project manager can be directly approached, but stakeholders can also drop their concern in a box (by means of a letter) that will be located just outside the management office. The box will also contain a message stating that the project management is obliged to take any type of concern seriously. The concerns will be documented and registered by the project manager or project coordinator. **Urgent concerns:** In case a concern need to be solved immediately (e.g. during project activities), the project manager or project coordinator can be directly contacted by the stakeholder. The project manager or coordinator will then try to find an immediate solution for the matter. The concern will be registered and documented by the project manager or coordinator afterwards. Also the solution that was used to solve the matter will be documented. **None urgent concerns:** In case the matter is not urgent (e.g. regarding planned activities or certain developments), the concern can be dropped in the message box outside the management office. The project manager can also be contacted directly by the stakeholder. Less urgent concerns will be discussed during the regular meetings (every 6 months) or an intermediate meeting will be arranged to discuss the matter. The project manager will negotiate the urgency of the matter with the stakeholder. Based on the urgency, a date and location will be arranged to discuss the matter. The concern will be documented by the project management. Afterwards, the solution that was used to solve the matter will be documented.

11. Evidence must be given that throughout the project activity any concerns by project stakeholders are documented and appropriately responded to by the management staff.

See point 10 above.

06 CO_2-fixation

The following pages are templates to document the parameters
- Future CO_2-fixation OR

Overview on the results of the analysis of the CO_2-fixation:

The table with management units has been removed from this template since all units have been uploaded to the website www.climateprojects.info. This has been done on the request of the pre-validator (see list with CAR's). All calculations and parameter settings can also be found in reference: 06_10_calculations_VER-futures.xls

Formula of calculation
Future CO_2-fixation [tCO_2/ha] = Future CO_2-fixation woody biomass [tCO_2] / Area of management unit [ha]

Selected carbon pools
- Aboveground woody b. = Stem volume * Biomass Expansion Factor * Wood density * Carbon fraction * C to CO_2 factor
- Belowground woody b. = Aboveground woody biomass * Root-to-Shoot ratio

Future CO_2-fixation

Management Unit(s)	LH1 (078); AH4 (080); CT11 (113); LH5 (118); SL3 (120) (9 ha) (***Alnus jorullensis***)	
Category of calculation: 2) Rotation forestry Time period until the Equilibrium Stand volume is reached: 25 years Timber (volume) grown during this time: 184 m3/ha	Inventories used to adapt the growth-model: The growth model has been derived from literature study (see Ref 06_01_Dunn_WW_1990.pdf, page 203, Chapter 2.3 Tree volume) and has not yet been validated/adapted with field data.	
Biomass Expansion Factor 1.1 Wood density 0.39 Carbon fraction 0.5 C to CO_2 factor 3.666 Root-to-Shoot ratio 0.23	Biomass Expansion Factor CFS default value Wood density International default value Root-to-Shoot ratio International default value	Reference: Not applicable Reference: Average value of various sources (see ref: 06_10_calculations_VER-futures.xls) Reference: 06_09_Wood_densities_IPCC_GL2006_Forest_Land.pdf, page 4.49, table 4.4, row 22 in third column (see also ref: 06_10_calculations_VER-futures.xls)
Justification of the growth-model	Source growth curves for Alnus jorullensis: Ref 06_01_Dunn_WW_1990.pdf, page 203, Chapter 2.3 Tree volume In this article a regression line for A. jorullenis (Volume per tree calculated by: 0,000841*AGE^2) was determined for similar plantation sites in Ecuador. Since the planting density will be 1480 trees/ha at the start of the project, the values per tree age were multiplied by 1480. Also thinnings are taken into account in the growth model. For the thinning the schedule proposed by Smit (1972) and Rojas et. al. (1978) will be applied (see 06_18_Lambrechts_1989.jpg, table 93, page 223)	
Future CO_2-fixation	176 tCO_2/ha	

This document will be made publicly available. CO_2-buyers will be able to use this information for marketing purposes.

Management Unit(s)	AH1 (068); AM2 (070); AM3 (071); CA7 (073); CH1 (074); CH2 (075); CO4 (076); AH5 (081); AH6; 082); AM5 (084); AM6 (085); CA3 (086); CH3 (087); CO1 (088); LH2 (091); LH4 (092); SL2 (096); AM1 (097); CA5 (099); CA6 (100); CA9 (101); CO2 (102); CO7 (104); CT7 (106); HU2 (109); LH3 (110); SC4 (112); SC2 (119); SL4 (121); CA1 (122); CT12 (126); CT15 (128); CT2 (129); CT8 (132); Hu4 (133); SC6 (134); SL5 (135) (128 ha) **(*Pinus patula*)**	
Category of calculation: 2) Rotation forestry Time period until the Equilibrium Stand volume is reached: 25 years Timber (volume) grown during this time: 210 m3/ha	Inventories used to adapt the growth-model: The growth model has been validated by PROFAFOR for the first 15 years (1993 – 2008) in plantations in a similar eco-region. For the remaining 10 years, the growth model was extrapolated. Methodology used by PROFAFOR: PROFAFOR methodology specifies the sampling intensity of 0.16% to monitor its plantation activities. This relates to one permanent sampling plot of 500 m2 for every 25 ha. With this sampling density the minimum precision target of +/- 10 %, at a 95% confidence level (t-value) for the mean standard deviation should be reached. The number of plots for each species is allocated based on the proportion of the area. In case it is necessary, the sample size is modified after the first monitoring event based on the variation of the carbon stock changes. Every permanent plot is marked and located with GPS coordinates.	
Biomass Expansion Factor 1.1 Wood density 0.43 Carbon fraction 0.5 C to CO_2 factor 3.666 Root-to-Shoot ratio 0.29	Biomass Expansion Factor CFS default value Wood density International default value Root-to-Shoot ratio International default value	Reference: Not applicable Reference: Average value of various sources (see ref: 06_10_calculations_VER-futures.xls) Reference: 06_09_Wood_densities_IPCC_GL2006_Forest_Land.pdf, page 4.49, table 4.4, row 15 in third column (see also ref: 06_10_calculations_VER-futures.xls)
Justification of the growth-model	**Source for P. patula:** PROFAFOR, Ecuador (2009) (See reference 06_17_Field data growth curves Pinus and Polylepis.msg). The data for P. patula, P. radiata and Polylepis incana is based on field measurements carried out in the plantations of PROFAFOR and has been validated up to 15 years. (The used tree density for all 3 tree species by PROFAFOR is 3X3m)	
Future CO_2-fixation	233 tCO_2/ha	

Management Unit(s)	AH3 (069); CA2 (072); CO5 (077); AH2 (079); AM4 (083); CO3 (089); CT1 (090); SC1 (093); SC5 (094); SL1 (095); CO6 (103); CT10 (105); CT9 (107); HU1 (108); SC3 (111); CT4 (116); HU3 (117); CA4 (124); CA8 (125); CT14 (127) (66 ha) **(*Pinus radiata*)**	
Category of calculation: 2) Rotation forestry Time period until the Equilibrium Stand volume is reached: 25 years Timber (volume) grown during this time: 402 m3/ha	Inventories used to adapt the growth-model: The growth model has been validated by PROFAFOR for the first 15 years (1993 – 2008) in plantations in a similar eco-region. For the remaining 10 years, the growth model was extrapolated. Methodology used by PROFAFOR: PROFAFOR methodology specifies the sampling intensity of 0.16% to monitor its plantation activities. This relates to one permanent sampling plot of 500 m2 for every 25 ha. With this sampling density the minimum precision target of +/- 10 %, at a 95% confidence level (t-value) for the mean standard deviation should be reached. The number of plots for each species is allocated based on the proportion of the area. In case it is necessary, the sample size is modified after the first monitoring event based on the variation of the carbon stock changes. Every permanent plot is marked and located with GPS coordinates.	
Biomass Expansion Factor 1.1 Wood density 0.41 Carbon fraction 0.5 C to CO_2 factor 3.666 Root-to-Shoot ratio 0.29	Biomass Expansion Factor CFS default value Wood density International default value Root-to-Shoot ratio International default value	Reference: Not applicable Reference: Average value of various sources (see ref: 06_10_calculations_VER-futures.xls) Reference: 06_09_Wood_densities_IPCC_GL2006_Forest_Land.pdf, page 4.49, table 4.4, row 15 in third column (see also ref: 06_10_calculations_VER-futures.xls)
Justification of the growth-model	**Source for P. radiata:** PROFAFOR, Ecuador (2009) (See reference 06_17_Field data growth curves Pinus and Polylepis.msg). The data for P. patula, P. radiata and Polylepis incana is based on field measurements carried out in the plantations of PROFAFOR and has been validated up to 15 years. (The used tree density for all 3 tree species by PROFAFOR is 3X3m)	
Future CO_2-fixation	432 tCO_2/ha	

Management Unit(s)	CT3 (115); CA10 (123); CT5 (130); CT6 (131) (10 ha) **(*Polylepis incana*)**	
Category of calculation: 1b) Conservation forest Time period until the Equilibrium Stand volume is reached: 25 years Timber (volume) grown during this time: 46 m3/ha	Inventories used to adapt the growth-model: The growth model has been validated by PROFAFOR for the first 15 years (1993 – 2008) in plantations in a similar eco-region. For the remaining 10 years, the growth model was extrapolated. Methodology used by PROFAFOR: PROFAFOR methodology specifies the sampling intensity of 0.16% to monitor its plantation activities. This relates to one permanent sampling plot of 500 m2 for every 25 ha. With this sampling density the minimum precision target of +/- 10 %, at a 95% confidence level (t-value) for the mean standard deviation should be reached. The number of plots for each species is allocated based on the proportion of the area. In case it is necessary, the sample size is modified after the first monitoring event based on the variation of the carbon stock changes. Every permanent plot is marked and located with GPS coordinates.	
Biomass Expansion Factor 1.1 Wood density 0.44 Carbon fraction 0.5 C to CO_2 factor 3.666 Root-to-Shoot ratio 0.46	Biomass Expansion Factor CFS default value Wood density International default value Root-to-Shoot ratio International default value	Reference: Not applicable Reference: see ref: 06_10_calculations_VER-futures.xls and/or 06_04_Wood_Density_Polylepis_Fehse_J_etal_2001.pdf, page 14, right text column below table 1) Reference: 06_09_Wood_densities_IPCC_GL2006_Forest_Land.pdf, page 4.49, table 4.4, column 3 ('Above ground biomass'), row 21. (see also ref: 06_10_calculations_VER-futures.xls)
Justification of the growth-model	**Source for P. incana:** PROFAFOR, Ecuador (2009) (See reference 06_17_Field data growth curves Pinus and Polylepis.msg). The data for P. patula, P. radiata and Polylepis incana is based on field measurements carried out in the plantations of PROFAFOR and has been validated up to 15 years. (The used tree density for all 3 tree species by PROFAFOR is 3X3m)	
Future CO_2-fixation	60 tCO_2/ha	

Formula of calculation
Baseline [tCO_2/ha] **=** (Baseline woody biomass [tCO_2] + Baseline non-woody biomass [tCO_2]) / Eligible planting area [ha]

08 Baseline

The following pages are templates to document the parameters

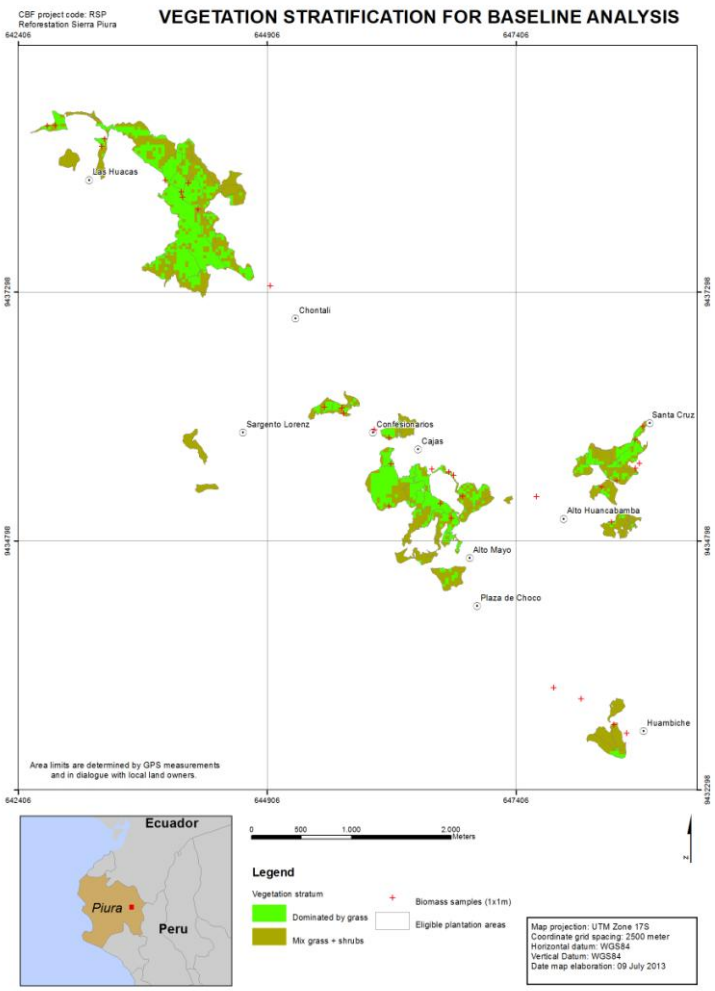
- Baseline woody biomass
- Results of the inventory on 'aboveground woody biomass'
- Baseline non-woody biomass

Please multiply the sheets according to the amount your project requires.

Describe shortly the process how the parameters were determined.

AREA STRATIFICATION

The project area is vegetated by two main vegetation types: Areas dominated by grass vegetation and areas dominated by a (complex) mixture of grasses and low shrubs (woody perennials). No cropland is present within the plantation areas. These 2 vegetation types were used for the baseline stratification of the area since it was expected that both vegetation types contained a considerable difference in biomass. Map 1 below shows the locations where both vegetation types are found in the project area (see also ref: 08_11_A4_Vegetation_stratification_for_baseline_analysis.pdf). The stratification of vegetation types was carried out with a Landsat ETM7 satellite image of 4 March 2011. A Maximum Likelihood Classification (MLC) algorithm was used in the classification procedure. Training data was collected in the field and used as input for the MLC. A total of 77 hectares of grass vegetation was found in the project area and 136 hectares of mixed vegetation (grass and shrubs).



Map with main vegetation types in the project area (Landsat ETM7 classification) (See ref: 08_11).

BASELINE: MOST ASUMABLE DEVELOPMENT OF THE PROJECT AREA (t = 25 years)

During the past decades, low technology agricultural practices, deforestation for firewood and extensive grazing has led to land degradation in the project area. Currently, the lands to be reforested within the project boundary are extensive grazing lands and former marginal agricultural lands. On these lands carbon stocks will continue to decrease or will remain stable. It is assumed that significant increases in the carbon stocks, in particular the living biomass pool of woody

perennials and the below-ground biomass of grasslands/shrubs, are not expected to occur in the absence of the project activity. As such, changes in carbon stocks shall be assumed to be zero.

EX-ANTE BASELINE FROM LITERATURE STUDY (t = 0 years)
From literature study, the following biomass values have been found for (degraded) grasslands in similar Andean ecozones at similar elevations:

Vegetation type	Value	Units	Description	Literature source
Andean grasslands, Central Ecuador, 3450 – 4000 masl	31-837 (≈ 3.2 tC/ha for above and below ground biomass))	g/m2	Above ground fresh biomass	08_09_Ramsay_2001_Biomass_Andes_Ecuador.pdf, page 163, chapter 'Aboveground standing biomass estimates'.
Undisturbed grassland, 3950 masl	2820 (≈ 11 tC/ha for above and below ground biomass)	g/m-2	Above ground fresh biomass	08_08_Hofstede_1995_Biomass_paramos_Colombia.pdf, page 6, table 2
Disturbed grassland, 3950 masl	868 (≈ 3.5 tC/ha for above and below ground biomass))	g/m-2	Above ground fresh biomass	08_08_Hofstede_1995_Biomass_paramos_Colombia.pdf, page 6, table 2

EX-ANTE BASELINE FROM FIELD SAMPLING (t = 0 years)
A total of 36 field samples have been taken from the two vegetation strata the plantation areas: 21 samples for the strata dominated by grass and 15 samples for the strata consisting of a mixture of grass and shrubs. Map 2 shows the locations where the samples were taken (see also ref: 08_05_A4_Overview_locations_biomass_samples.pdf). Samples were taken randomly for both vegetation types (1. Areas dominated by grass & 2. Areas dominated by a mixture grass/shrubs).



Map 2: Location of carbon samples (1x1m) in the project area (indicated as red crosses)

As indicated in map 2, the fresh biomass samples were taken at locations fairly evenly distributed over the plantation areas. A wooden frame of 1x1 meter was placed over the grassy vegetation. The vegetation was cut to ground level after which the entire amount the vegetation was weighted with a digital balance (see ref: 08_10_Field_work_biomass_sampling.doc for more details about the methodology).
In the table below, the result of the biomass measurements is shown:

Sample number	Reference to photo database	Fresh biomass weight (g/m2)	Vegetation class / strata
1	PM 1X1 AHBB.2	935	Mix of grass + shrubs
2	PM 1x1 Cajas	3265	Dominated by grass

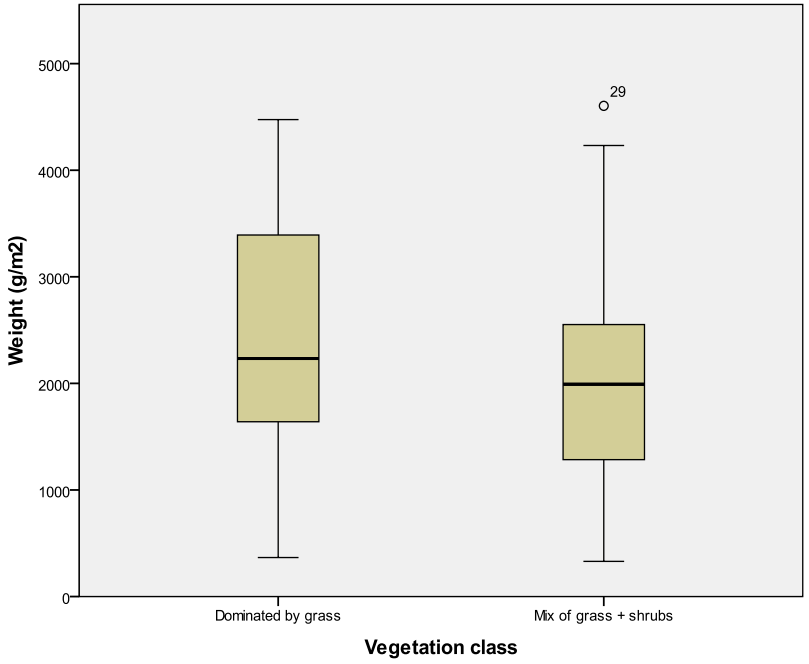
3	PM 1x1 Cajas.2	2291	Mix of grass + shrubs
4	PM 1X1 Cajas.3	1269	Dominated by grass
5	PM 1X1 Cajas.6	1162	Dominated by grass
6	PM 1X1 CHONTA	3392	Dominated by grass
7	PM 1X1 CHONTA.2	1993	Mix of grass + shrubs
8	PM 1X1 CHONTA.3	1640	Dominated by grass
9	PM 1X1 Confesion	4041	Dominated by grass
10	PM 1X1 Confesion.2	4232	Mix of grass + shrubs
11	PM 1 X1 Las Huacas	3851	Dominated by grass
12	PM 1X1 Las Huacas.2	2271	Mix of grass + shrubs
13	PM 1X1 Las Huacas.3	4475	Dominated by grass
14	PM 1X1 Las Huacas.4	4135	Dominated by grass
15	PM 1X1 Sta Cruz	1974	Mix of grass + shrubs
16	PM 1X1 AHBB	2444	Mix of grass + shrubs
17	PM 1X1 AHBB.3	4604	Mix of grass + shrubs
18	PM 1X1 Cajas.4	2400	Dominated by grass
19	PM 1X1 Cajas.5	1829	Dominated by grass
20	PM 1X1 Choco	2232	Dominated by grass
21	PM 1X1 CHONTA.4	2077	Dominated by grass
22	PM 1X1 Confesion.3	2598	Dominated by grass
23	PM 1X1 Huambiche	1921	Mix of grass + shrubs
24	PM 1X1 Sta Cruz.2	1964	Dominated by grass
25	PM 1X1 Sta Cruz.3	998	Dominated by grass
26	Cd1	2900	Mix of grass + shrubs
27	Cd2	3640	Dominated by grass
28	Cd3	3160	Dominated by grass
29	Cd4	1140	Mix of grass + shrubs
30	Cd5	2200	Dominated by grass
31	Cd6	2660	Mix of grass + shrubs
32	Cd7	530	Mix of grass + shrubs
33	Cd8	460	Dominated by grass
34	Cd9	330	Mix of grass + shrubs
35	Cd10	366	Dominated by grass
36	Cd11	1429	Mix of grass + shrubs

Table 1: Results of the fresh biomass sampling.

For each vegetation strata in table 1, the average values have been calculated (see table 2 below). See also the graph 1 for the maximum and minimum values that were found. In table 2, second column, we see that the average value for above ground fresh biomass content correspond fairly well with the value for undisturbed grassland found by R. Hofstede: 2820 g/m2 (on 3950masl) (see ref: 08_08_Hofstede_1995_Biomass_paramos_Colombia.pdf, page 6, table 2)

Vegetation class/strata	Average weight above fresh biomass (g/m2)
Dominated by grass	2436
Mix of grass + shrubs	2110

Table 2: Mean calculated values for fresh biomass per vegetation class/strata.



Graph 1: Boxplot graph with average values, maximum and minimum values per vegetation strata/class.

CONCLUSION BASELINE CALCULATION (t = 0 years)

The results of the baseline situation is shown in table 3 below. Each management unit can contain various vegetation strata and a management unit is rarely covered by only one type of vegetation. As such, the baseline was calculated for each individual management unit. Superficies of the strata *'grass'* and the strata *'grass + shrubs'* were calculated for each management unit. After that, the average biomass value measured per stratum was linked to the strata (see values in table 2). With the strata superficies and the average biomass value per stratum, the total amount of biomass was calculated for each individual management unit. The results are show in table 3. As can be seen, due to this methodology, the number of tCO2/ha varies for each individual management unit. No woody biomass was found in the project area. As such, the baseline for woody biomass is set to zero for all management units (and the entire project area).

Overview on the results of the baseline analysis:

ID (website)	ID of MU	Area (ha)	Eligible area (ha)	Baseline woody biomass (tCO_2/ha^1)	Baseline non-woody biomass (tCO_2/ha^2)
001	AH1	3	3	0	32
066	AH2	1	1	0	32

003	AH3	1	1	0	31
004	AH4	1	1	0	31
005	AH5	4	4	0	32
006	AH6	1	1	0	31
007	AM1	1	1	0	34
008	AM2	1	1	0	34
009	AM3	1	1	0	34
010	AM4	3	3	0	34
011	AM5	1	1	0	31
012	AM6	2	2	0	32
013	AM7	1	1	0	31
014	CA1	2	2	0	35
015	CA10	3	3	0	33
059	CA2	2	2	0	34
025	CA3	4	4	0	35
026	CA4	14	14	0	34
016	CA5	6	6	0	33
017	CA6	1	1	0	36
018	CA7	2	2	0	35
019	CA8	1	1	0	34
020	CA9	3	3	0	34
021	CH1	3	3	0	31
022	CH2	5	5	0	32
067	CH3	5	5	0	32
027	CO1	2	2	0	33
028	CO2	1	1	0	33
029	CO3	1	1	0	34
030	CO4	2	2	0	34
031	CO5	1	1	0	33
060	CO6	3	3	0	31

This document will be made publicly available. CO_2buyers will be able to use this information for marketing purposes.

034	CO7	1	1	0	32
035	CT1	7	7	0	33
036	CT10	4	4	0	35
037	CT11	2	2	0	31
038	CT12	2	2	0	31
039	CT13	1	1	0	32
040	CT14	2	2	0	31
041	CT15	14	14	0	33
061	CT2	20	20	0	35
043	CT3	1	1	0	34
044	CT4	7	7	0	33
062	CT5	4	4	0	35
046	CT6	2	2	0	34
047	CT7	4	4	0	34
048	CT8	2	2	0	32
033	CT9	3	3	0	33
049	HU1	2	2	0	31
053	HU2	2	2	0	31
064	HU3	2	2	0	31
052	HU4	3	3	0	32
053	LH1	1	1	0	31
054	LH2	3	3	0	34
055	LH3	3	3	0	32
065	LH4	1	1	0	34
057	LH5	3	3	0	31
058	SC1	3	3	0	33
059	SC2	1	1	0	34
060	SC3	4	4	0	34
061	SC4	2	2	0	32
062	SC5	2	2	0	33

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063	SC6	2	2	0	33
064	SL1	3	3	0	31
065	SL2	4	4	0	32
066	SL3	1	1	0	31
067	SL4	8	8	0	34
068	SL5	5	5	0	32

Table 3: Results of the baseline situation for each Management Unit.

[1] No woody biomass is found in the plantation areas.
[2] Each individual management unit can contain various vegetation strata. As such, the unit *tCO2/ha* has not been calculated for the vegetation strata but for each individual management unit instead. This approach also corresponds with the approach on the ClimateProjects website (www.climateprojects.info), where total carbon calculations are carried out on management unit level. See also ref: 08_03_Calculations_of_Baseline_non_woody_v4.xls for the calculations in the table above.

Baseline non-woody biomass

Formula of calculation
Baseline non-woody biomass [tCO_2]
= (Baseline aboveground non-woody biomass [tCO_2] + Baseline belowground non-woody biomass [tCO_2]) * Eligible planting area [ha]

Selected carbon pools
- Aboveground non-woody biomass = Fresh biomass * Dry-to-Wet ratio * Carbon fraction * C to CO_2 factor
- Belowground non-woody biomass = Aboveground non-woody biomass * Root-to-Shoot ratio

Stratum	Dominated by grass (77 ha)	
Study used 21 field samples (1x1m) + literature study or Inventory use, see references: 08_03_Calculations_of_Baseline_non_woody_v4.xls 08_05_A4_Overview_locations_biomass_samples.pdf	Result: 12.2 tons of dry biomass per ha	
Biomass 24.4 tons / ha (wet biomass) Wet-to-Dry ratio 0.5 Carbon fraction 0.5 C to CO_2 factor 3.666 Root-to-Shoot ratio 0.61	Fresh biomass Local default value Wet-to-Dry ratio CFS default value Root-to-Shoot ratio CFS default value	Reference: 08_03_Calculations_of_Baseline_non_woody_v4.xls Reference: 08_03_Calculations_of_Baseline_non_woody_v4.xls Reference: CarbonFix standard
Baseline non-woody biomass	36 tCO_2/ha	

Stratum	Dominated by grass + shrubs (136 ha)
Study used 15 field samples (1x1m) + literature study or Inventory use, see references: 08_03_Calculations_of_Baseline_non_woody_v4.xls 08_05_A4_Overview_locations_biomass_samples.pdf	Result: 10.6 tons of dry biomass per ha

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Biomass Wet-to-Dry ratio Carbon fraction C to CO_2 factor Root-to-Shoot ratio	21.1 tons / ha (wet biomass) 0.5 0.5 3.666 0.61	Fresh biomass Local default value Wet-to-Dry ratio CFS default value Root-to-Shoot ratio CFS default value	Reference: 08_03_Calculations_of_Baseline_non_woody_v4.xls Reference: 08_03_Calculations_of_Baseline_non_woody_v4.xls Reference: CarbonFix standard
Baseline non-woody biomass		31 tCO_2/ha	

Formula of calculation
Leakage [tCO_2/ha] = Leakage aboveground woody biomass [tCO_2] / Eligible planting area project area [ha]

09 Leakage

The following pages are templates to document the parameters

- Leakage aboveground woody biomass - category a. to e.
- Leakage aboveground woody biomass - category f.

Please multiply the sheets according to the amount your project requires.

Describe shortly the process how the parameters were determined.
All the parameters below are based on stakeholder surveys that were carried out in 2009. The Peruvian project developers are very familiar with the customs of the rural community (Communidad Yamango) in the project area and the problems that this community faces. **Overall conclusion for Leakage:** The project area where reforestation activities are planned are currently not being used for any agricultural activities, nor for other types of landuse: The area is abandoned and not being used for any kind of CO2 emitting activities (firewood collection, intensive animal grazing, etc.) that will be displaced by the project activities. Evidence can be provided to the certifier during field visits that there is no displacement and that the lands surrounding the project activity contain no significant biomass. Also evidence can be provided that these lands are likely to receive the shifted activities. Hence, leakage can be considered zero.

Describe shortly the selection of your categories.

a. fuel wood use	Firewood consumption in the rural Andes regions is not homogeneous. It varies with the climate conditions y with the diet of the people. As such, it has been observed that in the colder parts of the Andes (over 2500 masl), farmers keep the fireplace burning until late at night to warm themselves. This doesn't happen in the warmer (lower) parts of the Andes. Also the custom to prepare 'Menestre' at lunch (a traditional meal) is demanding more energy than preparing for example rice, Yuca or potato. We have calculated that this behaviour is requiring around 1.5m3 of firewood on a monthly basis for a family of around 5 persons (around 160kg/family). This value is supported by a research carried out in the Ecuadorian High Andes that found an average firewood consumption of 1.01m3/month for a family of 5-7 persons (see ref 09_04_S_Wunder_1996_Firewooduse_Ecuador.pdf, page 376, table 12). According to C.E Van Dam (1986:7), firewood collection would be sustainable with an average of 0.3 and 3 ha. of forest per family (depending on ecology, soil quality and tree species).   Photo1 Photo 2 Photo 1 y Photo 2: Cooking with firewood in Choco. Since the project area doesn't contain woody biomass that is suitable for firewood, farmers have to walk about 1-1,5 hours to find firewood (see interviews on videos that are uploaded on the CarbonFix website). Some of the farmers go to the native forest 'Bosque Mijal', that is located about 1.5-2 km in the North-West of the plantation area. Hence, there will not be any leakage for firewood due to the project activities since farmers do not collect firewood in the project area >> Leakage = 0

b. charcoal burning	The farmers in the project area do not use or make charcoal. No leakage will be caused by displacement of charcoal burning activities >> Leakage = 0
c. harvesting of timber	Quality timber is being used for construction, fencing, road construction, tree nurseries, tools, furniture, etc.  Photo 3. Timber use for contruction  Photo 4. Nurcery fence in Alto Mayo Since quality timber doesn't grow in the project area suitable for construction timber, farmers have to walk about 1-1,5 hours to find the nearest timber sources (see interviews on videos uploaded on the carbonfix website). Some of the farmers go to the nearby forest 'Bosque Mijal', that is located about 1.5-2 km in the North-West of the plantation area. Hence, there will not be any leakage for (construction) timber due to the project activities since farmers do not collect timber in the project area >> Leakage = 0
d. agriculture farming	The plantation areas have been appointed by the farmers/landowners in the area. No agricultural areas are present within the plantation areas. As such no agricultural activities will have to be displaced because of the project activities >> Leakage = 0
e. resettlement	The plantation areas have been appointed by the farmers/landowners in the area. No settlements are present within the plantation areas. As such, no settlements will be displaced because of the planned project activities >> Leakage = 0

f. livestock grazing	The majority of the families in the project area do not keep cattle, because of the low productivity of the pastures. The pastures have been invaded by weeds, such as Selcique, which make the reforestation areas less suitable for livestock grazing. The cattle that is present in the area is in bad condition and do not have a (high) commercial value. As such, the cattle and other animals that are present in the area are for subsistence only. Some farmers allow their animals to graze on private properties (with permission of the owner) where the pastures are of better quality. However, these pastures are located outside the project area. The cattle in the area consist of some sheep, very few cows and horses. According to the interviews held in October and November 2009, only a small percentage of the families keep animals. Those families have an average of 4 animals. The UNPCC methodology for baseline and leakage calculations (see ref: 09_01_UNPCCC_methodology_baseline_and_Leakage.doc, chapter IV, page 8, point 29) mentions the following: "*If the area of the cropland within the project boundary displaced due to the project activity is lower than 10 per cent of the total project area, and the number of domesticated grazing animals displaced is less than 10% of the average grazing capacity (see appendix D for calculations) of the project area, and the time-average number of domesticated roaming animals displaced is less than 10% of the average grazing capacity per hectare (see appendix D for calculations) of the project area, then: Leakage = 0*" In the supplementary documentation the grazing capacity (head/ha) of the project area has been calculated for sheep, using default international values (see ref: 09_02_Calculation_leakage.xls). According to this calculation, the grazing capacity of the project area is around 3 head/ha (for sheep) which is results in a total average GC of 213ha * 3 = 639 heads. The number of animals that will be displaced by the reforestation activities are less than 70 (<10%). As such, leakage for livestock grazing will be zero.

General Terms & Conditions

(Version 3.1)

Parties

- CarbonFix e.V., Friedrichstr. 15, 70174 Stuttgart, Germany (*CarbonFix*)
- You (*User*)

Recitals

- These General Terms & Conditions (GTC) supplement the CarbonFix Standard (CFS) and set further criteria for its usage.
- The *User* wishes to use the CFS in accordance with the rules of the CFS and these GTC as amended from time to time (see section 6).

Content

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1. Definitions

1.1. Every word that is underlined with dashes is defined in the part 'Terms' of the CFS.
1.2. *CarbonFix* stands for the association as well as for its representatives, employees or agents.
1.3. *User* can be any project developer as well as CO2-buyer.

2. Commencement, amendment and termination of these GTC

2.1. These GTC commence on the date on which the *User* indicates its acceptance.
2.2. *CarbonFix* may amend these GTC in accordance with the chapter 'Adaption of the CFS' of the CFS.
2.3. They shall continue in effect until terminated by finishing the respective relationship between the *User* and *CarbonFix*.

3. General

3.1. The *User* declares/agrees to the following:
   a. The accuracy of the information submitted to *CarbonFix*.
   b. In respect to the CFS, to comply with all national and international laws concerned + that (potential) conflicts will be announced to *CarbonFix* immediately.
   c. To communicate in a fast and clear manner with *CarbonFix* in any matter concerning the project as well as the CFS.
   d. Every project can be visited, upon prior notice, by CO2-buyers as well as interested people in order to receive a better impression on how the project is set-up. Costs are born by the visitor.
   e. That CO2-certificates must not and will not be double-used in any way.

4. Concerning project developers

4.1. The *User* declares/agrees to the following:
   a. All project information submitted via the ClimateProjects website may be used by *CarbonFix* as well as the CO2-buyers of the project.
   b. The project may be inspected unannounced by *CarbonFix* to verify the project's information. Costs for such unannounced inspections are carried by *CarbonFix*. It is still expected that the inspectors will be assisted by the project developer in its on-site logistics and the organization of accommodation.
   c. Pre-validation fees or any other costs relating to that process are not refunded in case of a rejection.
   d. Sales fees or any other costs relating to that process are not refunded in case of an exclusion.
   e. Every sale agreement with a CO2-buyer must respect the rules and regulations of the CFS and these GTC.

5. Concerning CO2-buyers

5.1. By purchasing CO2-certificates the *User* declares/agrees to the following:
   a. In case of a project exclusion in accordance with the chapter 'Project exclusion' of the CFS, to accept the consequences and not file suit against *CarbonFix*.
   b. Any legal consequences of a project exclusion have to and will be carried out between the contractual parties.

6. Privacy and User Information

6.1. The *User* allows *CarbonFix* to use its personal information for administrative use as well as for sending relevant information, including the newsletter.

6.2. CarbonFix stores all information of the User that is necessary for running the CFS most efficiently.

6.3. *CarbonFix* will not allow third parties access to the *User* information without its prior approval, unless *CarbonFix* is required to do so by law.

6.4. The *User* may request *CarbonFix* to access and correct its personal information at any time.

7. Intellectual Property / Copyright

7.1. The CFS including its name, logo, texts and images are copyrighted property of *CarbonFix*. All rights reserved. The communication package as referred to in the chapter 'Communication policy' of the CFS applies.

7.2. Each party acknowledges and agrees that, except for the rights expressly provided for in the CFS, these GTC and any rights otherwise agreed upon between the parties, neither party shall acquire any rights, title or interest in or to any (pre-)existing Intellectual Property Rights, including but not limited to name and logo of the other party.

8. Liability and Limitation

8.1. CarbonFix does not give any representation, warranty or guarantee with the CFS documents. *CarbonFix* shall not be liable for any mistakes or damages resulting from the use of this information unless it is based on gross negligence or intention. This limitation does not apply to damages to life, body or health.

8.2. *CarbonFix* expressively disclaims any liability for losses due to project rejections or exclusions.

8.3. *CarbonFix* is not responsible for any actions or omissions of any of the project participants.

8.4. The compensation of lost CO2-certificates by the CFS buffer (as outlined in the chapter 'CFS Buffer') will be provided to the extent possible. There is no title for being compensated by the buffer. CFS is not liable in any way for the functioning or a lack of compensation by the buffer.

9. Indemnity

9.1. To the extent permissible by law, the *User* indemnifies *CarbonFix* from and against all proceedings, actions, claims, demands, losses, liabilities, damages, costs and expenses which may be made or brought against or suffered or incurred by using the CFS.

9.2. The indemnities in these GTC are continuing obligations of the *User*, separate and independent from other obligations and survive the termination of these GTC. Furthermore they are absolute and unconditional and unaffected by anything that might have the effect of prejudicing, releasing, discharging or affecting in any other way the liability of the party giving the indemnity.

10. Legal Disputes

10.1. German Law is applicable for any legal disputes with *CarbonFix*. The place of jurisdiction is Staufen, Germany (Registrar of Association: Amtsgericht Staufen No. VR 429).

11. Relationship between the Parties

11.1. Unless otherwise stated, *CarbonFix* is permitted by the *User* to use its name (including logo) for all purposes concerning the CFS.

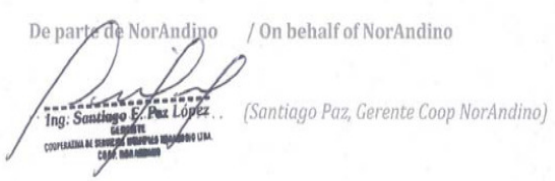
11.2. Nothing in these GTC will be taken to create a relationship between the parties of agency, joint venture, fiduciary relationship, partnership or other joint undertaking.

12. Procedural

12.1. Any unclear or unsolved issue shall be dissolved by the technical board of *CarbonFix*.
12.2. These GTC shall not be overruled by opposing rules or regulations of the addressed *User*.
12.3. Rules between an individual *User* and *CarbonFix* that are different to the ones as set out in these GTC are only valid if they are in written form.
12.4. The rights and obligations herein are personal to both parties and may not be assigned or delegated to a third party without the express written consent of the other party.
12.5. If individual terms of these GTC are invalid or unenforceable the validity of other terms shall not be affected.

Name of the Project	Reforestation Sierra Piura
Version of the Standard	CarbonFix Standar v3.0

I have read, understood and agree to the GTC of *CarbonFix*.



Place, Date 15/11/2010 Signature (upload scanned signature)